FS Migration Validation Engine

Microsoft Founders Hub Application Pack

Technical Master Document with Diagrams

Version 2.0 - June 2026

Status: Ready for Review

Based on: FS Migration Validation Engine v1.4

Target Program: Microsoft for Startups Founders Hub

Classification: Confidential

Table of Contents

Phase 1 - Azure Founders Hub Pack

Phase 1 - Azure Founders Hub Pack (v4 Cloud-Ready Edition)

Version: 2.0

Based on: FS Migration Validation Engine v1.4

Target Program: Microsoft for Startups Founders Hub

Status: Ready for Review

Executive Summary

This pack presents the FS Migration Validation Engine -- an automated, metadata-driven data migration validation platform purpose-built for regulated Financial Services institutions.

After extensive technical analysis (Parts 1-4) covering legacy system assessment, current architecture evaluation, gap analysis, and migration strategy, we have transitioned from research into productization. This documentation pack represents the complete technical, commercial, and strategic foundation for Microsoft Founders Hub application.

Product Positioning

> ***An Automated Data Migration Validation & Governance Platform for Regulated Financial Services.***

The platform is no longer just a data migration tool. It has evolved into an enterprise-grade Migration Assurance Platform that automates legacy system discovery, intelligent schema mapping, migration validation, governance, scoring, and compliance auditing for regulated financial data migrations.

The Five Platform Pillars

Pillar	Description	Status
--------	-------------	--------
1. Enterprise Discovery	Automated schema discovery, me	? Built
2. Intelligent Mapping	Schema comparison, similarity	? Built
3. Migration Validation	10 structured controls, rule e	? Built
4. Governance & Audit	Version control, audit history	? Built
5. AI Intelligence Layer	AI-assisted mapping recommenda	? In Development

Document Inventory

#	Document	Audience	Purpose
---	----------	----------	---------
1.1	Product Overview	Microsoft reviewers, investors	Executive summary, problem/sol
1.2	Technical Architecture	Architecture reviewers, engine	Platform architecture, compone

1.3	Platform Core Definition	Architecture reviewers	Orchestration layer, event bus
1.4	Azure Cloud Architecture	Microsoft reviewers, cloud arc	Azure service mapping, deploym
1.5	Azure Reference Architecture D	Technical reviewers	Visual architecture diagram, s
1.6	Technical Narrative	Founders Hub, investors	Full narrative: problem, solut

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Evidence Portfolio

All claims in this pack are backed by working code and technical analysis:

Evidence	Reference	Status
----------	-----------	--------
? Legacy Assessment	Part 1 Analysis	Complete
? Current Architecture Assessm	Part 3 Analysis	Complete
? Migration Strategy	Part 4 Analysis	Complete
? Working CLI Engine	`app/main.py` -- 10 controls, s	Complete
? FastAPI Endpoints	`app/api/` -- REST API layer	Complete
? Database Schema	`sql/schema/` -- PostgreSQL eng	Complete
? Docker Deployment	`Dockerfile`, `docker-compose.	Complete
? Governance Framework	Release gates, control depende	Complete
? Gap Analysis	Phase 1.3 NOT_REQUIRED doc	Complete
? Security Scanning	`bandit-report.html`, `trivy-f	Complete

Target Market

Primary: Financial Services -- Banks (Tier 1-3), Insurers, Asset Managers, FinTech, Regulated Enterprises

Secondary: Any organisation undertaking regulated data migration (Healthcare, Government, Utilities)

The platform is designed for organisations where data integrity, compliance, and auditability during migration are non-negotiable requirements.

Program Alignment

This documentation pack is designed for:

- ? Microsoft for Startups Founders Hub -- Primary target
- ? AWS Activate -- Adaptable with minor cloud service mapping changes
- ? Google for Startups -- Adaptable with minor cloud service mapping changes
- ? Investor Due Diligence -- Angel, Seed, Series A
- ? Enterprise Customer Evaluations -- Technical and commercial reviews

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Key Differentiators

Feature	Our Platform	Manual Approach	Legacy ETL Tools
---------	-------------	-----------------	------------------

Automated schema discovery	? Automated	? Manual analysis	? Requires pre-configuration
10 structured validation contr	? Built-in	? Ad-hoc testing	?? Partial
Confidence scoring	? Automated	? Subjective	? Not available
Audit trail & governance	? Built-in	? Manual documentation	?? Limited
Release gate enforcement	? Automated	? Manual approval	? Not available
Control dependency management	? Built-in	? Manual sequencing	? Not available
Docker deployment	? Ready	N/A	?? Legacy installs
Financial Services focus	? Purpose-built	? Generic	? Generic

Next Steps

Following this master overview, review the documents in order:

1. Phase 1.1 -- Product Overview (non-technical executive summary)
2. Phase 1.2 -- Technical Architecture (engineering deep-dive)
3. Phase 1.3 -- Platform Core Definition (orchestration architecture)
4. Phase 1.4 -- Azure Cloud Architecture (cloud-native migration plan)
5. Phase 1.5 -- Azure Reference Architecture Diagram (visual architecture)
6. Phase 1.6 -- Technical Narrative (complete Founders Hub narrative)
7. Phase 2.1-2.10 -- Commercial & Investor Readiness Package

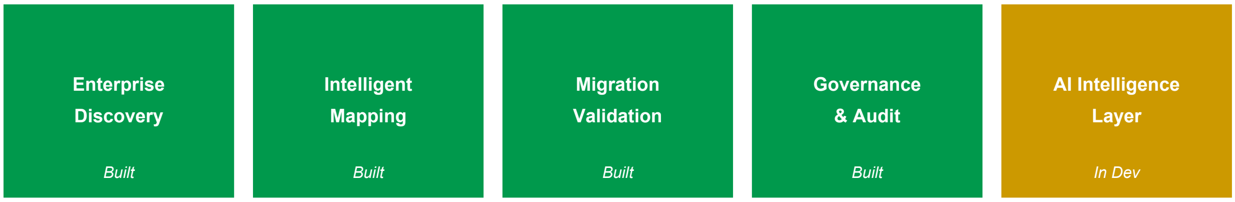

Figure 1: The Five Platform Pillars

Phase 1.1 - Product Overview

Phase 1.1 -- Product Overview

Version: 2.0

Product: FS Migration Validation Engine

Target: Microsoft for Startups Founders Hub

Audience: Non-technical reviewers, investors, enterprise customers

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Executive Summary

Financial Services institutions undertaking data migration projects face a fundamental challenge: how to ensure that millions of records move from legacy systems to modern platforms accurately, completely, and in compliance with regulatory requirements.

Traditional approaches rely on manual sampling, ad-hoc spreadsheet tracking, and post-migration firefighting. This is slow, error-prone, and unacceptable in regulated environments where data integrity failures can trigger regulatory penalties, operational losses, and reputational damage.

The FS Migration Validation Engine solves this by providing an automated, metadata-driven validation platform that systematically verifies every aspect of a data migration -- from schema compatibility and data type correctness through to referential integrity, business rule compliance, and regulatory reporting accuracy.

The Problem

Data migration in Financial Services is high-risk and highly regulated:

Challenge	Impact
-----------	--------
Manual validation	~60% of validation effort is m
Incomplete coverage	Sampling-based checks miss edg
Poor auditability	Spreadsheet-driven tracking fa
No repeatability	Every migration project starts
High cost of failure	Data errors can cause operatio
Lack of governance	No formal release gates, appro

Banks migrating core banking systems, payment platforms, or regulatory reporting systems cannot afford data quality failures. Yet most migration validation today relies on custom scripts, manual checks, and hope.

Our Solution

The FS Migration Validation Engine is an automated, metadata-driven validation and governance platform that:

1. Discovers legacy and target system schemas automatically
2. Validates data across 10 structured migration controls
3. Scores migration quality with objective, repeatable metrics
4. Governs the migration lifecycle with release gates and audit trails
5. Reports comprehensive audit evidence suitable for regulatory review

How It Works

```
Legacy System ??-> Discovery ??-> Validation Controls ??-> Scoring ??-> Audit Report
                                         ?
                                  Release Gate Review
                                         ?
                                Migration Approved / Blocked
```

The platform runs as a CLI engine, REST API, or Docker container -- integrating into existing CI/CD pipelines and deployment workflows.

Platform Capabilities

1. Automated Discovery

- Connect to source and target databases
- Discover schemas, tables, columns, and relationships
- Extract metadata for validation planning
- Identify schema differences automatically

2. Structured Validation Controls (C01-C010)

Control	Description
---------	-------------
C01	Row count reconciliation
C02	Column-level data type validat
C03	Nullability constraint verific
C04	Primary key integrity check
C05	Foreign key referential integr
C06	Business rule validation
C07	Date/time boundary validation
C08	Numeric range and precision ch
C09	String pattern and format vali
C010	Regulatory field completeness

3. Scoring & Release Gates

- Each control receives an objective pass/fail score
- Aggregate scoring provides migration quality metrics
- Configurable release gates block migrations below minimum quality thresholds
- Dependency management ensures controls execute in correct order

4. Governance & Audit

- Every control execution is logged with timestamp and result
- Audit-ready CSV exports for regulatory review
- Batch-level tracking with unique batch IDs
- Exception register for all validation failures
- Full execution history for compliance reporting

5. REST API

- FastAPI-based RESTful interface
- Run validations programmatically
- Integrate with CI/CD pipelines
- Query results and export audits remotely

Business Benefits

Benefit	Measurable Impact
---------	------------------
Reduced manual effort	70-80% reduction in manual val
Complete coverage	100% of records validated, not
Regulatory compliance	Audit-ready evidence for every
Repeatable process	Standardised validation across
Faster migrations	Automated gates replace manual
Lower risk	Early detection of data qualit
Governance by design	Release gates prevent bad data

Current Development Status

Component	Status
-----------	--------
CLI Engine (10 controls, scori	? Production-ready
REST API layer	? Built (FastAPI)
Database schema (PostgreSQL)	? Production-ready
Docker deployment	? Ready
Control dependency DAG	? Implemented
Release gate enforcement	? Implemented

Audit export (CSV)	? Implemented
Security scanning (bandit, tri	? Integrated
Web dashboard	? In development
AI-assisted mapping recommenda	? Planned
Multi-tenant SaaS deployment	? Planned

Why Azure

Microsoft Azure provides the enterprise-grade foundation required for regulated Financial Services deployments:

- Azure Container Apps -- Scalable, serverless compute for the validation engine
- Azure SQL Database -- Managed relational database with high availability
- Azure Blob Storage -- Secure audit report and evidence storage
- Azure Key Vault -- Encrypted secrets management for database credentials
- Microsoft Entra ID -- Enterprise identity and role-based access control
- Azure Monitor -- Centralised logging and performance monitoring
- Azure API Management -- Secure API gateway for enterprise integration

The platform is designed to be Azure-native while supporting hybrid and on-premises deployment for customers with specific regulatory requirements.

Product Vision

Our vision is to become the standard platform for regulated data migration assurance -- providing Financial Services institutions with the automation, governance, and auditability they need to migrate mission-critical data with confidence.

Future capabilities include:

- AI-assisted mapping recommendations -- Using schema analysis to suggest optimal field mappings
- Intelligent mapping engine -- Automated schema matching with confidence scoring
- Interactive dashboards -- Real-time migration quality visualisation
- Multi-cloud support -- Extend validation to AWS, GCP, and hybrid deployments
- Regulatory template library -- Pre-built controls for specific regulations (Basel, SOX, FCA, PRA)

Competitive Advantages

Area	Our Advantage
------	--------------
Purpose-built	Designed specifically for regu
10 structured controls	Comprehensive validation frame
Governance-first	Release gates, audit trails, a
Automated scoring	Objective, repeatable quality
Docker-native	Deploy anywhere -- cloud, on-pr
API-first	Integrate with existing CI/CD

Open, extensible	Python-based, open architectur

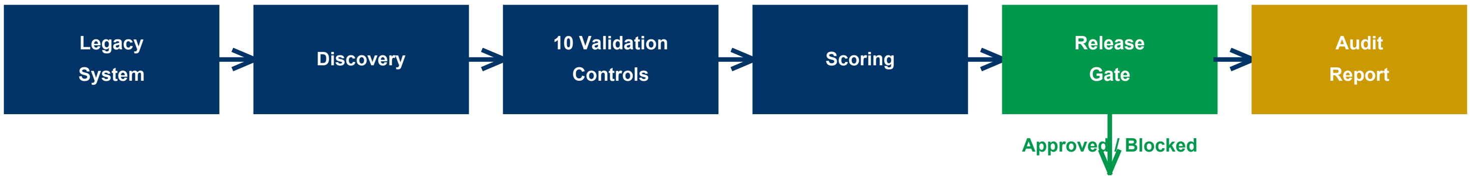

Figure 2: How It Works - Legacy System to Audit Report Flow

Phase 1.2 - Technical Architecture

Phase 1.2 -- Technical Architecture

Version: 2.0

Product: FS Migration Validation Engine

Status: Architecture Baseline

Audience: Microsoft Founders Hub, Technical Reviewers, Engineering Team

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1. Introduction

The FS Migration Validation Engine is an automated, metadata-driven data migration validation and governance platform designed for regulated Financial Services environments. It validates data migrations across core banking, payments, lending, asset management, and regulatory reporting systems.

The platform is architected as a modular, cloud-ready solution with a centralised orchestration layer (Platform Core) that coordinates independent domain engines responsible for discovery, validation, scoring, and governance.

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2. Architectural Principles

Principle	Description
-----------	-------------
Modularity	Independent engines communicat
Metadata-driven	All validation is driven by da
Security-by-design	Zero Trust, encryption at rest
Governance-first	Release gates, audit trails, a
Extensibility	Plugin-based engine architectu
Cloud-native	Designed for Azure, deployable
Observability	Comprehensive logging, metrics
Repeatability	Deterministic validation -- sam

---

3. High-Level Architecture

```
                    Users / CLI / API
                          ?
                   ???????????????
                   ?  API Layer  ?
                   ?  (FastAPI)  ?
                   ???????????????
                          ?
                   ???????????????
                   ? Platform   ?
                   ? Core       ?
                   ? (Orchestrator)?
                   ???????????????
                          ?
        ?????????????????????????????????????
        ?                 ?                 ?
  ?????????????   ???????????????   ???????????????
  ? Discovery ?   ?  Validation ?   ?  Scoring    ?
  ? Engine    ?   ?  Engine     ?   ?  Engine     ?
  ?????????????   ???????????????   ???????????????
        ?                 ?                 ?
        ?????????????????????????????????????
                          ?
                   ???????????????
                   ? Governance  ?
                   ? Engine      ?
                   ???????????????
                          ?
                   ???????????????
                   ?  Database   ?
                   ?  (PostgreSQL)?
                   ???????????????
```

4. Core Platform Components

4.1 Platform Core

The central orchestration layer responsible for:

- Workflow orchestration -- Controlling execution sequence of discovery, validation, scoring
- Control dependency management -- DAG-based execution (C02 depends on C01, C09 depends on C03)
- State management -- Tracking batch execution state, checkpointing
- Configuration management -- Centralised config for all engines
- Security enforcement -- Authentication, authorisation, audit logging
- Event routing -- Emitting and consuming events between engines
- Error handling -- Failure isolation, retry logic, graceful degradation

4.2 Discovery Engine

Responsible for automated schema discovery from source and target databases.

Capabilities:

- Database connection management via app/db_connector.py
- Schema discovery -- tables, columns, data types, constraints
- Metadata extraction and repository population
- Relationship detection (primary keys, foreign keys)

Output: Metadata repository in the engine database.

4.3 Validation Engine

The core validation component executing 10 structured migration controls.

Validation Controls (C01-C010):

Control	Description	SQL Template
---------	-------------	-------------
C01	Row count reconciliation	`sql/controls/`
C02	Column-level data type validat	`sql/controls/`
C03	Nullability constraint verific	`sql/controls/`
C04	Primary key integrity check	`sql/controls/`
C05	Foreign key referential integr	`sql/controls/`
C06	Business rule validation	`sql/controls/`
C07	Date/time boundary validation	`sql/controls/`
C08	Numeric range and precision ch	`sql/controls/`
C09	String pattern and format vali	`sql/controls/`
C010	Regulatory field completeness	`sql/controls/`

Controls are implemented as SQL templates executed against source and target databases, with results stored in the engine database.

4.4 Scoring Engine

Responsible for evaluating validation results and producing quality metrics.

Capabilities:

- Control-level pass/fail determination
- Aggregate scoring across all controls
- Configurable pass thresholds
- Weighted scoring by control criticality
- Exception register management

4.5 Governance Engine

Provides enterprise governance across the migration lifecycle.

Capabilities:

- Release gate enforcement -- blocks migration if score below threshold
- Batch-level tracking with unique batch IDs
- Audit trail -- every execution logged with timestamp and result
- Exception register -- all validation failures recorded
- CSV audit export for regulatory review

---

5. Data Architecture

5.1 Engine Database (PostgreSQL)

The engine database stores all operational data:

Entity	Description
--------	-------------
`migration_validation_batch`	Batch-level execution tracking
`migration_control_execution`	Individual control execution r
`migration_exception_register`	All validation exceptions
Source/target metadata	Discovered schema information

See sql/schema/01_engine_schema.sql for full schema definition.

5.2 Source & Target Databases

The platform connects to source and target PostgreSQL databases to perform validation. The connector layer (app/db_connector.py) manages connections and handles credential security through environment variables and Azure Key Vault.

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6. Security Architecture

Layer	Implementation
-------	----------------
Identity	Environment-based authenticati
Secrets	Environment variables -> Azure
Encryption at rest	PostgreSQL TDE / Azure SQL tra
Encryption in transit	TLS for all database connectio
API security	FastAPI with CORS, future: OAu
Secrets isolation	No credentials in source code
Audit logging	Every control execution logged

---

7. Integration Architecture

7.1 CLI Interface

```
python app/main.py run --config config.yaml
python app/main.py discover --config config.yaml
python app/main.py export --config config.yaml --batch-id <id>
```

7.2 REST API (FastAPI)

- GET /health -- Health check
- POST /api/v1/validate -- Run validation batch
- GET /api/v1/batch/{id} -- Get batch results
- GET /api/v1/export/{batch_id} -- Export audit CSV

7.3 Docker Deployment

- docker-compose up --build -- Full stack deployment
- Containerised engine, database, and API

---

8. Deployment Architecture

Environment	Deployment Model	Database
-------------	-----------------	----------
Development	Local Docker Compose	Local PostgreSQL
Test/QA	Azure Container Apps	Azure SQL Database
Production	Azure Container Apps / AKS	Azure SQL Database
On-premises	Docker Compose	PostgreSQL

9. Current Implementation Status

Component	Status	Location
-----------	--------	----------
CLI Engine	? Production-ready	`app/main.py`, `app/execution_
10 Validation Controls	? Implemented	`sql/controls/`
Scoring Engine	? Implemented	`app/scoring_engine.py`
Governance / Release Gates	? Implemented	`config.yaml`, `app/execution_
Audit Export	? Implemented	`app/audit_export.py`
Database Schema	? Production-ready	`sql/schema/`
Docker Deployment	? Ready	`Dockerfile`, `docker-compose.
REST API	? Built	`app/api/`
Discovery Service	? Built	`app/services/dataset_discover
Rule Executor	? Built	`app/rule_executor.py`, `app/r
Web Dashboard	? In Development	`dashboard/`
AI Mapping Engine	? Planned	`app/mapping_engine/`
Multi-tenant SaaS	? Planned	Future phase

10. Technology Stack

Layer	Technology
-------	-----------
Language	Python 3.11
API Framework	FastAPI, Uvicorn
Database	PostgreSQL 15+
Containerisation	Docker, Docker Compose
Configuration	YAML + Environment Variables
Validation	SQL-driven control templates
Reporting	CSV export (future: interactiv
Security	python-jose, cryptography
Testing	pytest

11. Scalability

The architecture supports horizontal scaling through:

- Stateless API -- FastAPI instances can be scaled horizontally
- Independent engine scaling -- Each engine can scale independently
- Database connections -- Connection pooling via PostgreSQL
- Container orchestration -- Kubernetes-ready (Docker images)
- Async processing -- Future: background task queues for large batches

---

12. Future Architecture Evolution

```
Current:                  Future:
????????????            ????????????
? CLI +    ?            ? Web UI   ?
? API      ?            ? + API    ?
????????????            ????????????
? Platform ?            ? Platform ?
? Core     ?            ? Core     ?
????????????            ????????????
? Validation?           ? Discovery?
? Controls  ?           ? Mapping  ?
?           ?           ? Validation?
?           ?           ? Scoring  ?
?           ?           ? AI       ?
?           ?           ? Governance?
????????????            ????????????
```

The modular Platform Core architecture ensures that new engines can be added without architectural redesign. The AI Intelligence Layer (Azure OpenAI) can be integrated as an additional engine providing mapping recommendations, anomaly detection, and natural language querying.

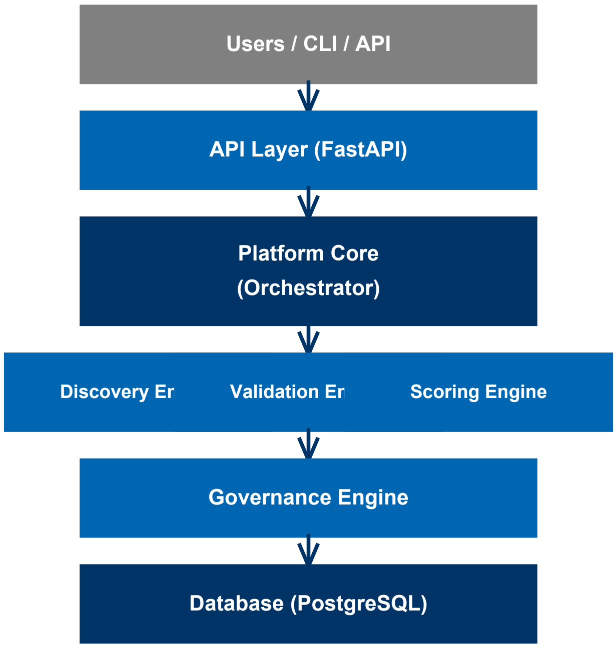

Figure 3: Platform Architecture Diagram

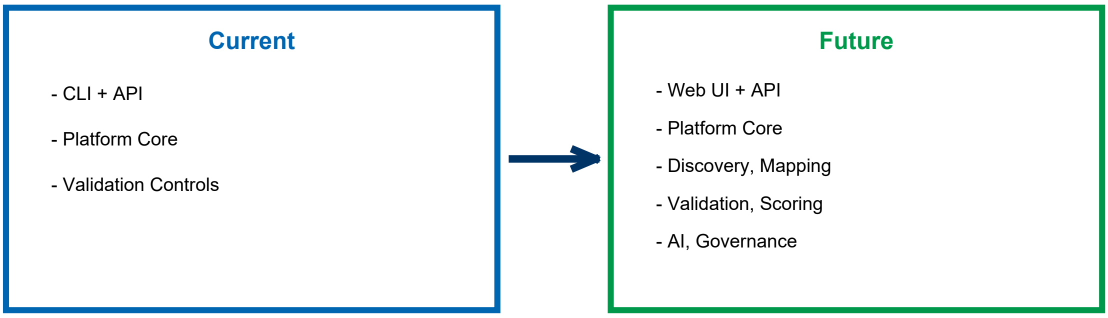

Figure 4: Architecture Evolution - Current to Future

Phase 1.3 - Platform Core Definition

Phase 1.3 -- Platform Core Definition

Version: 2.0

Status: Architecture Baseline

Audience: Engineering, Architecture, Founders Hub Technical Reviewers

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Platform Core

Executive Summary

The Platform Core is the central orchestration layer of the FS Migration Validation Engine. Rather than allowing each engine to communicate directly with every other engine, the Platform Core acts as the single coordination point responsible for workflow orchestration, state management, event routing, security enforcement, configuration management, and engine lifecycle management.

This architecture dramatically reduces coupling between components while improving scalability, maintainability, security, and extensibility.

Why the Platform Core Exists

Without a Platform Core:

```
Discovery ? Validation ? Scoring ? Governance ? Export
```

Every engine develops dependencies on every other engine, creating duplicated logic, inconsistent state, difficult testing, tight coupling, and deployment challenges.

With a Platform Core:

```
              Users / CLI / API
                     ?
             Platform Core
                     ?
     ?????????????????????????????????
     ?       ?       ?       ?       ?
Discovery Validation Scoring Governance Export
```

Each engine communicates only with the Platform Core. No engine requires knowledge of another engine's internal implementation.

Core Responsibilities

1. Workflow Orchestration

Coordinates execution between engines in the correct sequence:

```
Discovery -> Platform Core -> Validation -> Platform Core -> Scoring -> Platform Core -> Gover
```

The engines remain independent. The Platform Core determines execution order.

2. Control Dependency Management

Manages the Directed Acyclic Graph (DAG) of control dependencies:

```
C01
 ?
 ???-> C02
      ?
      ???-> C03 -> C09
                ?
                ???-> C04 -> C05 -> C06 -> C07 -> C08 -> C010
```

Dependencies are defined in config.yaml:

```
control_dependencies:
  C02: [C01]
  C09: [C03]
```

3. Engine Registry

Maintains metadata for every registered engine:

Engine	Status	Version	Capabilities
--------	--------	---------	-------------
Discovery	Healthy	1.0	Schema discovery, metadata ext
Validation	Healthy	1.0	10 structured controls (C01-C0
Scoring	Healthy	1.0	Pass/fail, aggregate scoring
Governance	Healthy	1.0	Release gates, audit, export

4. State Management

The Platform Core owns workflow state:

```
Pipeline Status: Running
??? Discovery: Complete (passed 100%)
??? Validation: Running (5/10 controls complete)
??? Scoring: Pending
??? Governance: Pending
```

This provides restart capability, checkpointing, resilience, and recovery.

5. Configuration Service

Central configuration management via config.yaml:

```
execution:
  environment: DEV
  client_name: DemoBank
  control_id: null        # null = run all controls

release_gate:
  enabled: true
  block_on_status: ["FAIL", "BLOCKED", "ERROR"]
  minimum_score: 80
  enforcement_mode: "STRICT"

rules:
  C01: enabled
  C02: enabled
  # ... through to C010
```

Every engine reads configuration from one location -- the Platform Core.

6. Security Enforcement

The Platform Core validates every request:

- Authentication (API keys, environment-based)
- Authorisation (role-based execution permissions)
- Audit logging (every action logged with batch ID and timestamp)
- Secrets isolation (credentials via environment variables, not code)
- Configuration validation (pre-execution checks)

7. Error Handling & Failure Isolation

- Control-level isolation -- A failure in C01 does not block C03
- Graceful degradation -- Non-critical failures logged, execution continues
- Retry logic -- Configurable timeout per control (default: 300 seconds)
- Exception register -- All failures recorded for audit
- Recovery mode -- Re-run only failed controls from previous batch

8. Observability

Provides a single operational view:

Metric	Source
--------	--------
Engine execution time	Platform Core timing
Control pass/fail rates	Scoring engine results
Batch duration	Workflow start/end timestamps
Exception counts	Exception register
API latency	FastAPI middleware

Interaction Model

```
User / CLI
    ?
    ?
API Gateway (FastAPI)
    ?
    ?
Authentication
    ?
    ?
Platform Core
    ?
    ???-> Workflow Manager
    ?        ?
    ?        ???-> Discovery Engine
    ?        ?        ?
    ?        ?        ???-> Platform Core (event)
    ?        ?
    ?        ???-> Validation Engine
    ?        ?        ?
    ?        ?        ???-> Control C01
    ?        ?        ???-> Control C02
    ?        ?        ???-> ...
    ?        ?        ???-> Control C010
    ?        ?                 ?
    ?        ?                 ???-> Platform Core (event)
    ?        ?
    ?        ???-> Scoring Engine
    ?        ?        ?
    ?        ?        ???-> Platform Core (event)
    ?        ?
    ?        ???-> Governance Engine
    ?        ?        ?
    ?        ?        ???-> Release Gate Check
    ?        ?        ?
    ?        ?        ???-> Platform Core (event)
    ?        ?
    ?        ???-> Export Engine
    ?                 ?
    ?                 ???-> Audit CSV
    ?
    ?
Response (pass/fail, score, exceptions, audit)
```

No engine communicates directly with another. All coordination passes through the Platform Core.

Current Implementation

The Platform Core is implemented in app/execution_engine.py:

```
class ExecutionEngine:
    def __init__(self, config, batch_id=None):
        self.config = config
        self.batch_id = batch_id or str(uuid.uuid4())
        self.recovery_mode = False

    def run(self):
        # 1. Initialise batch in database
        # 2. Resolve control execution order (DAG)
        # 3. Execute each control in dependency order
        # 4. Score results
        # 5. Apply release gates
        # 6. Export audit trail
```

Key files:

- app/execution_engine.py -- Core orchestrator
- app/rule_executor.py -- Individual control execution
- app/rule_factory.py -- Control instantiation from config
- app/scoring_engine.py -- Scoring logic
- app/audit_export.py -- CSV audit export
- config.yaml -- Centralised configuration
- app/db_connector.py -- Database connectivity

Azure Alignment

The Platform Core maps naturally to Azure services:

Platform Core Function	Azure Service
------------------------	---------------
Compute / Hosting	Azure Container Apps
API Gateway	Azure API Management
Identity	Microsoft Entra ID
Secrets	Azure Key Vault
Monitoring	Azure Monitor / Application In
Database	Azure SQL Database / PostgreSQ
Storage	Azure Blob Storage
CI/CD	GitHub Actions / Azure DevOps
Infrastructure	Bicep / ARM Templates

Architectural Principles

1. Loose coupling -- Engines communicate only through the Platform Core
2. High cohesion -- Each engine has a single, well-defined responsibility
3. Deterministic execution -- Same inputs produce same outputs
4. Stateless engines -- State managed by Platform Core
5. Centralised governance -- Release gates, audit, and compliance enforced centrally

6. Cloud-native -- Designed for Azure, deployable anywhere
7. Extensibility -- New engines can be added without modifying existing ones
8. Security-by-design -- Authentication, authorisation, and audit built into the core
9. Failure isolation -- One control failure does not cascade
10. Observability-by-default -- Every execution produces audit evidence

---

Benefits

Technical

- Reduced coupling between components
- Easier maintenance and testing
- Independent engine evolution
- Simplified debugging (single coordination point)
- Improved resilience and recovery

Operational

- Centralised monitoring and alerting
- Consistent configuration management
- Unified security enforcement
- Simplified deployment model
- Faster troubleshooting

Business

- Faster feature delivery (add engines without redesign)
- Lower maintenance costs
- Easier partner integration
- Enterprise-grade governance built-in
- Clear Azure alignment for cloud migration

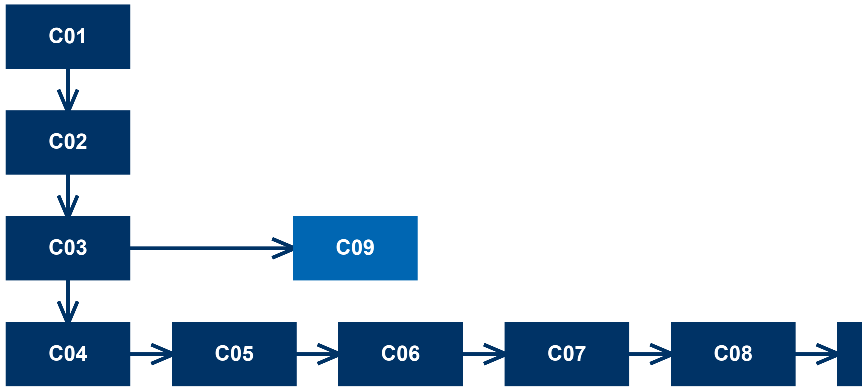

Figure 5: Control Dependency DAG - C01 to C010 Execution Order

Phase 1.4 - Azure Cloud Architecture

Phase 1.4 -- Azure Cloud Architecture

Version: 2.0

Product: FS Migration Validation Engine

Status: Founders Hub Ready

Audience: Microsoft Reviewers, Cloud Architects, Engineering Team

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Executive Summary

The FS Migration Validation Engine is designed as an Azure-native, cloud-first SaaS solution built on Microsoft's Well-Architected Framework. It leverages managed Azure services to maximise scalability, security, reliability, and operational efficiency for regulated Financial Services data migration validation.

The architecture follows a Platform Core + microservices model where independent domain engines (Discovery, Validation, Scoring, Governance) are orchestrated through a central coordination layer. The solution is cloud-portable while taking full advantage of Azure's managed services ecosystem.

High-Level Architecture

```
                  Users / CLI / DevOps
                         ?
            ?????????????????????????????
            ?                           ?
         Web Portal                  REST API
         (Future)                    (FastAPI)
            ?                           ?
            ?????????????????????????????
                         ?
                Azure Front Door (Optional)
                         ?
                 Azure API Management
                         ?
              Microsoft Entra ID (Future)
                         ?
                   Platform Core
                         ?
     ?????????????????????????????????????????????
     ?          ?          ?          ?          ?
 Discovery  Validation  Scoring   Governance  Reports
  Engine     Engine     Engine     Engine     Engine
     ?          ?          ?          ?          ?
     ?????????????????????????????????????????????
                         ?
              Azure SQL / PostgreSQL
                         ?
                 Azure Blob Storage
```

Core Azure Services

Azure Container Apps (Recommended MVP)

The primary compute platform for the validation engine.

Advantages:

- Serverless container hosting
- Automatic scaling based on validation workload
- Minimal operational overhead
- Cost-efficient consumption pricing
- Native Dapr support for future microservices
- Fast deployment from Docker images

Migration path: Azure Kubernetes Service (AKS) for enterprise-scale deployments.

Azure API Management

Secure gateway for all API interactions.

Responsibilities:

- API gateway for REST endpoints
- Rate limiting and throttling
- API versioning
- Authentication enforcement (future: OAuth2 / Entra ID)

- Request validation
- Usage analytics
- Developer portal for integration documentation

Microsoft Entra ID (Future Integration)

Identity and access management for enterprise deployments.

Capabilities:

- Single Sign-On (SSO)
- OAuth2 / OpenID Connect
- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)
- Conditional Access policies
- Enterprise identity federation

Azure SQL Database / PostgreSQL Flexible Server

Primary data store for the validation engine.

Stores:

- Migration batch records
- Control execution results
- Exception register
- Audit trail
- Configuration metadata
- Customer and project data (future)

Benefits:

- Managed service with automated backups
- High availability with zone redundancy
- Built-in threat detection (Defender for Cloud)
- Encryption at rest and in transit

Azure Blob Storage

Secure storage for audit artifacts and exports.

Stores:

- Audit CSV exports
- Validation reports
- Uploaded schema definitions
- Log archives
- Backup snapshots

Benefits:

- Cost-effective tiered storage
- Lifecycle management policies
- Immutable storage for compliance
- Geo-redundant replication

Azure Key Vault

Centralised secrets management.

Stores:

- Database connection strings
- API keys
- Certificates
- Encryption keys
- Environment credentials

Security:

- No secrets in source code or configuration files
- Managed identities for service-to-service authentication
- Access policies with least privilege
- Audit logging for all secret access

Azure Monitor & Application Insights

Comprehensive observability.

Captures:

- Infrastructure metrics (CPU, memory, network)
- API response times and error rates
- Control execution duration
- Batch processing performance
- Exception tracking
- Distributed tracing across microservices
- Custom dashboards and alerts

GitHub Actions / Azure DevOps

CI/CD pipeline automation.

Pipeline stages:

```
Commit -> Build -> Unit Tests -> Security Scan -> Container Build -> Deploy to Test -> Integra
```

Infrastructure as Code:

- Bicep / ARM Templates for Azure resource provisioning
- Version-controlled infrastructure definitions
- Reproducible deployments across environments

Networking Architecture

```
Internet
    ?
    ?
Azure Front Door (Optional)
    ?
    ?
Azure API Management
    ?
    ?
Virtual Network
    ?
    ??? Azure Container Apps (Private Endpoints)
    ??? Azure SQL Database (Private Endpoint)
    ??? Azure Key Vault (Private Endpoint)
    ??? Azure Blob Storage (Private Endpoint)
```

Security features:

- Private networking for all data services
- Network Security Groups (NSGs)
- DDoS protection
- TLS 1.2+ for all communications
- Managed certificates via Azure Key Vault
- Service endpoints / Private Link for PaaS services

Security Architecture

The platform follows a Zero Trust security model:

Security Layer	Implementation
----------------	----------------
Identity	Microsoft Entra ID + RBAC (fut
Authentication	API keys -> OAuth2 / Entra ID (
Secrets	Azure Key Vault, no secrets in
Network	Private endpoints, TLS, NSGs
Data at rest	Azure SQL TDE, Blob encryption
Data in transit	TLS 1.2+ for all connections
Monitoring	Defender for Cloud, Sentinel (
Auditing	Full audit trail for every exe

Deployment Environments

Environment	Compute	Database	Purpose
-------------	---------	----------	---------
Development	Local Docker	Local PostgreSQL	Local development and testing
Test/QA	Azure Container Apps	Azure SQL (shared)	Automated testing, CI/CD
Staging	Azure Container Apps	Azure SQL (isolated)	Pre-production validation
Production	Azure Container Apps (MVP) -> A	Azure SQL (HA)	Customer deployments

---

Scalability Strategy

Each engine scales independently based on workload:

Engine	Scale Trigger
--------	--------------
API Layer	Concurrent API requests
Validation Engine	Number of controls x batch siz
Scoring Engine	Post-validation scoring reques
Audit Export	Export request volume

Autoscaling is managed by Azure Container Apps using:

- CPU utilisation thresholds
- Memory utilisation thresholds
- HTTP request queue depth
- Custom scaling rules (future)

High Availability

Azure services provide enterprise-grade availability:

- Zone-redundant deployment (Azure Container Apps)
- Active geo-replication (Azure SQL Database)
- Automatic failover with managed services
- Automated backups with point-in-time restore
- Health probes for container instances
- Rolling updates for zero-downtime deployments

Azure Well-Architected Framework Alignment

Reliability

- Managed Azure services with built-in redundancy
- Automated health monitoring and recovery
- Fault-tolerant architecture (failure isolation per control)
- Automated backups and point-in-time restore

Security

- Zero Trust architecture
- Microsoft Entra ID + Azure Key Vault
- Private endpoints for data services
- Defender for Cloud continuous assessment
- Full audit trail for compliance

Cost Optimisation

- Consumption-based pricing (Azure Container Apps)
- Autoscaling to match workload demand
- Managed PaaS services reduce operational overhead
- Blob storage lifecycle policies for cost-effective archiving
- Right-sizing through Azure Advisor recommendations

Operational Excellence

- CI/CD with GitHub Actions / Azure DevOps
- Infrastructure as Code (Bicep / ARM)
- Centralised monitoring (Azure Monitor)
- Automated deployments with quality gates
- Comprehensive audit logging

Performance Efficiency

- Stateless API design for horizontal scaling
- Independent engine scaling
- Connection pooling for database efficiency
- Background processing for long-running validations
- Caching strategies for metadata (future)

Azure Service Mapping

Platform Capability	Azure Service
---------------------	---------------
Compute (MVP)	Azure Container Apps
Compute (Enterprise)	Azure Kubernetes Service (AKS)
API Gateway	Azure API Management
Global Routing	Azure Front Door
Identity	Microsoft Entra ID
Primary Database	Azure SQL Database / PostgreSQ
Secrets Management	Azure Key Vault
Object Storage	Azure Blob Storage
Monitoring	Azure Monitor
Application Telemetry	Application Insights
Security Posture	Microsoft Defender for Cloud
CI/CD	GitHub Actions / Azure DevOps
Infrastructure as Code	Bicep / ARM Templates

Future Azure Services

Phase	Service	Purpose

-------	---------	---------
Phase 2	Azure OpenAI Service	AI-assisted mapping recommenda
Phase 2	Azure Service Bus	Async event processing for lar
Phase 3	Azure Cosmos DB	Global-scale metadata storage
Phase 3	Azure Cognitive Search	Semantic search across mapping
Phase 3	Power BI	Interactive dashboards and rep
Phase 4	Microsoft Fabric	Enterprise data integration

Cost Model (Projected)

Service	Estimated Monthly Cost (MVP)
---------	-----------------------------
Azure Container Apps	$200-500
Azure SQL Database (Basic)	$150-300
Azure Blob Storage	$50-100
Azure API Management	$100-200
Azure Key Vault	$10-20
Azure Monitor	$50-100
Total (MVP)	**$560-1,220/month**

Costs scale linearly with customer adoption. Enterprise deployments with AKS and HA configurations range from $2,000-5,000/month.

Conclusion

This Azure-native architecture provides a secure, scalable, and cost-effective foundation for the FS Migration Validation Engine. By combining a centralised Platform Core with independently deployable microservices and managed Azure services, the platform minimises operational overhead while maximising resilience, extensibility, and enterprise readiness for regulated Financial Services environments.

The design supports rapid MVP deployment through Azure Container Apps while providing a clear migration path to Azure Kubernetes Service for large-scale enterprise deployments. It establishes a robust foundation for future AI capabilities, multi-tenant SaaS operations, and a growing ecosystem of platform integrations.

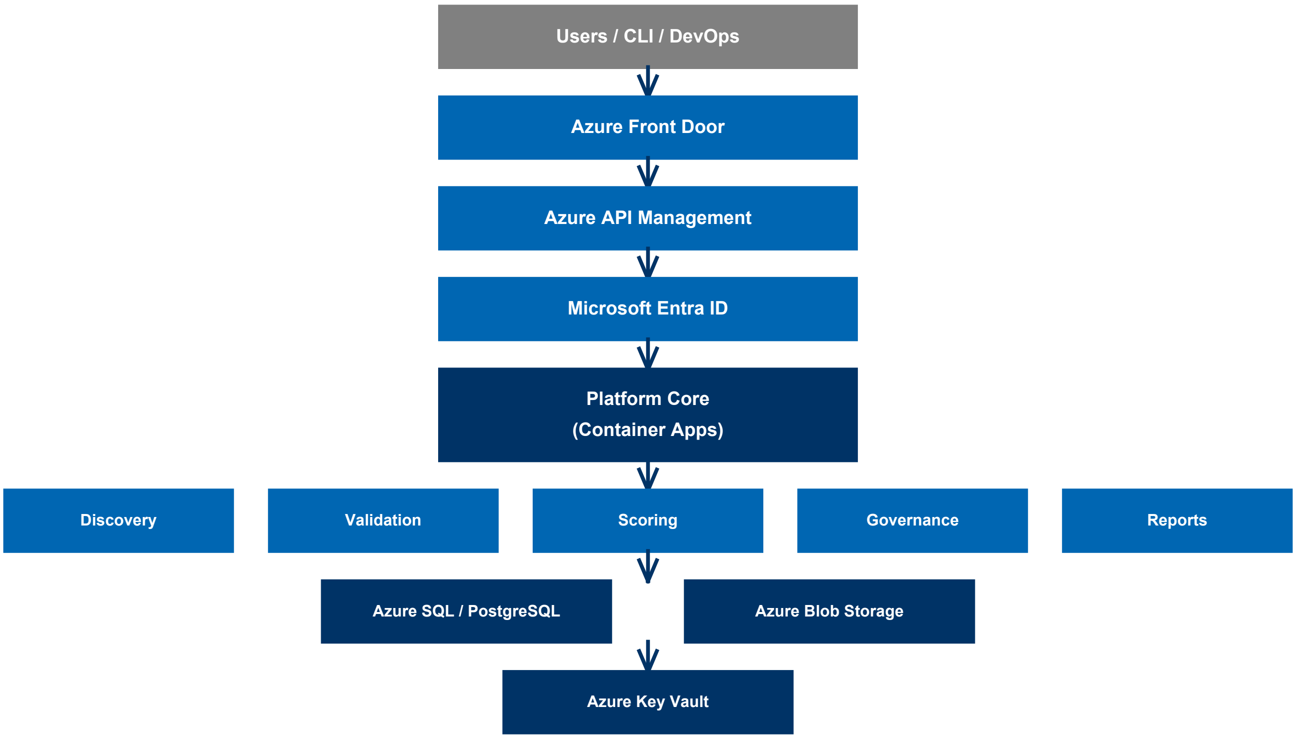

Figure 6: Azure Cloud Architecture - Service Integration

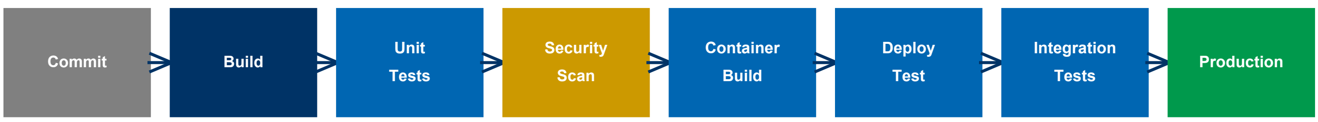

Figure 7: CI/CD Pipeline Stages

Phase 1.5 - Azure Reference Architecture Diagram

Phase 1.5 -- Azure Reference Architecture Diagram

Version: 2.0

Architecture Style: Azure-Native - Microservices - Event-Driven - Financial Services Grade

Product: FS Migration Validation Engine

Azure Reference Architecture

```
                                    ????????????????????????????????????
                                    ?           USERS / CLI            ?
                                    ?  Developers - DevOps - Auditors  ?
                                    ????????????????????????????????????
                                                   ?
                                        HTTPS / REST API / OAuth2
                                                   ?
                     ????????????????????????????????????????????????????????????
                     ?                Azure Front Door (Optional)               ?
                     ?         Global Routing - WAF - DDoS - SSL Offload        ?
                     ????????????????????????????????????????????????????????????
                                                ?
                                   ????????????????????????????????
                                   ?   Azure API Management       ?
                                   ?   API Gateway - Rate Limits   ?
                                   ?   Versioning - Policies       ?
                                   ????????????????????????????????
                                                  ?
                                   ????????????????????????????????
                                   ?   Microsoft Entra ID         ?
                                   ?   OAuth2 - SSO - MFA - RBAC  ?
                                   ????????????????????????????????
                                                  ?
????????????????????????????????????????????????????????????????????????????????????
                         PLATFORM CORE (Azure Container Apps)
????????????????????????????????????????????????????????????????????????????????????

                          ???????????????????????????????????????
                          ?       Platform Core API             ?
                          ?-------------------------------------?
                          ?  Workflow Orchestration             ?
                          ?  Control Dependency (DAG) Manager   ?
                          ?  Configuration Service              ?
                          ?  Security Enforcement               ?
                          ?  State Management                   ?
                          ?  Audit Routing                      ?
                          ???????????????????????????????????????
                                         ?
        ???????????????????????????????????????????????????????????????????
        ?                                ?                                ?
        ?                                ?                                ?
????????????????????          ????????????????????              ????????????????????
? Discovery Engine ?          ? Validation Engine?              ?  Scoring Engine  ?
? Schema Discovery ?          ? C01 - C010       ?              ? Pass/Fail Scoring?
? Metadata Extract ?          ? SQL Controls     ?              ? Aggregate Metrics?
????????????????????          ????????????????????              ????????????????????
          ?                             ?                                 ?
          ?????????????????????????????????????????????????????????????????
                               ?                    ?
                    ????????????????????   ????????????????????
                    ? Governance Engine?   ?   Export Engine  ?
                    ? Release Gates    ?   ?  Audit CSV       ?
                    ? Exception Register?   ?  Batch Reports   ?
                    ????????????????????   ????????????????????
                             ?                      ?
                             ????????????????????????
                                        ?
????????????????????????????????????????????????????????????????????????????????????
                           DATA & STORAGE LAYER
????????????????????????????????????????????????????????????????????????????????????

    ?????????????????????      ??????????????????????      ?????????????????????
    ?  Azure SQL /      ?      ?  Azure Blob        ?      ?  Azure Key Vault  ?
    ?  PostgreSQL       ?      ?  Storage           ?      ?                   ?
    ?????????????????????      ??????????????????????      ?????????????????????
    ?  Batch Records    ?      ?  Audit CSVs        ?      ?  DB Credentials   ?
    ?  Control Results  ?      ?  Reports           ?      ?  API Keys         ?
    ?  Exception Register?     ?  Logs              ?      ?  Certificates     ?
```

```
     ?  Schema Metadata  ?      ?  Backups           ?      ?  Encryption Keys  ?
     ?????????????????????      ??????????????????????      ?????????????????????
                         ?
 ????????????????????????????????????????????????????????????????????????????????????
                             MONITORING & OBSERVABILITY
 ????????????????????????????????????????????????????????????????????????????????????

                     ??????????????????????????????????????
                     ?  Azure Monitor + Application       ?
                     ?  Insights                          ?
                     ??????????????????????????????????????
                     ?  Infrastructure Metrics            ?
                     ?  API Performance                   ?
                     ?  Control Execution Duration        ?
                     ?  Exception Tracking                ?
                     ?  Distributed Tracing               ?
                     ?  Custom Dashboards                 ?
                     ??????????????????????????????????????

 ????????????????????????????????????????????????????????????????????????????????????
                                CI/CD & DEVOPS
 ????????????????????????????????????????????????????????????????????????????????????

                     ??????????????????????????????????????
                     ?  GitHub Actions / Azure DevOps     ?
                     ??????????????????????????????????????
                     ?  Build -> Test -> Security Scan      ?
                     ?  -> Container Build -> Deploy        ?
                     ??????????????????????????????????????
                     ?  Infrastructure as Code            ?
                     ?  (Bicep / ARM Templates)           ?
                     ??????????????????????????????????????
```

Architectural Layers

1. Presentation Layer

- CLI Interface -- python app/main.py run --config config.yaml
- REST API -- FastAPI endpoints for programmatic access
- Future Web Portal -- Interactive dashboard and management UI

2. API Layer

- Azure API Management -- Gateway, rate limiting, versioning
- Microsoft Entra ID -- Authentication and authorisation (future)
- Azure Front Door -- Global routing and WAF (optional)

3. Platform Core

- Workflow orchestration -- Coordinates engine execution sequence
- Control dependency management -- DAG-based execution ordering
- Configuration management -- Centralised via config.yaml
- Security enforcement -- Authentication, audit, secrets isolation
- State management -- Batch tracking, checkpointing, recovery
- Error handling -- Failure isolation, retry, graceful degradation

4. Domain Engines

Each engine is independently deployable and scalable:

Engine	Function	Status
--------	----------	--------
Discovery Engine	Schema discovery, metadata ext	? Built
Validation Engine	10 structured controls (C01-C0	? Built
Scoring Engine	Pass/fail, aggregate scoring	? Built
Governance Engine	Release gates, exception regis	? Built
Export Engine	CSV audit export	? Built

5. Data Layer

- Azure SQL / PostgreSQL -- Primary operational database
- Azure Blob Storage -- Audit artifacts and report storage
- Azure Key Vault -- Secrets and credentials management

6. Observability Layer

- Azure Monitor -- Infrastructure and platform metrics
- Application Insights -- Application performance monitoring
- Custom dashboards -- Migration quality visualisation

7. DevOps Layer

- GitHub Actions / Azure DevOps -- CI/CD pipeline automation
- Bicep / ARM Templates -- Infrastructure as Code
- Automated testing -- Unit tests, integration tests, security scans

Deployment Model

Environment	Compute	Database	Storage	Configuration
-------------	---------	----------	---------	---------------
Development	Local Docker	Local PostgreSQL	Local filesystem	`config.yaml` + `.env`
Test	Azure Container Apps	Azure SQL (DTU)	Blob Storage (Cool)	Key Vault + Config
Staging	Azure Container Apps	Azure SQL (vCore)	Blob Storage (Cool)	Key Vault + Config
Production	AKS (future)	Azure SQL (HA)	Blob Storage (Hot)	Key Vault + Config

Security Boundaries

Boundary	Control
----------	---------
Network	Private endpoints for all data
Identity	Environment-based -> Microsoft

Authentication	API keys -> OAuth2 (migration p
Secrets	Azure Key Vault -- no secrets i
Data at rest	Azure SQL TDE, Blob encryption
Data in transit	TLS 1.2+ for all communication
Audit	Every execution logged with ba

Scalability Model

Component	Scale Strategy	Metric
-----------	---------------	--------
API Layer	Horizontal (Container Apps)	Requests/second
Validation Engine	Horizontal	Controls x batch size
Scoring Engine	Vertical	Result volume
governance Engine	Singleton	Batch completion events
Database	Vertical -> Read replicas	Connection pool depth

Future Expansion

The architecture is designed to support:

- Phase 2: AI-assisted mapping (Azure OpenAI), async processing (Service Bus)
- Phase 3: Web dashboard (React + Azure Static Web Apps), multi-tenant SaaS
- Phase 4: Marketplace, partner integrations, Microsoft Fabric connectivity

Phase 1.6 - Azure Founders Hub Technical Narrative

Phase 1.6 -- Azure Founders Hub Technical Narrative

Version: 2.0

Program: Microsoft for Startups Founders Hub

Product: FS Migration Validation Engine

Executive Summary

Our platform is an automated, metadata-driven data migration validation and governance platform purpose-built for regulated Financial Services institutions. It systematically validates data migrations across core banking, payments, lending, asset management, and regulatory reporting systems -- ensuring accuracy, completeness, and regulatory compliance.

The solution is designed as an Azure-native, cloud-first platform that combines Microsoft's managed cloud services with intelligent automation to reduce the cost, complexity, and risk of financial data migrations. By adopting managed Azure services, a modular Platform Core architecture, and metadata-driven validation, we enable financial institutions to validate migrations with complete coverage, objective scoring, and auditable governance.

Business Problem

Financial Services institutions undertaking data migration projects face critical challenges:

Challenge	Impact
-----------	--------
Regulatory scrutiny	FCA, PRA, Basel, and SOX requi
Manual validation	60%+ of validation effort is m
Incomplete coverage	Sampling-based checks miss edg
No repeatability	Every migration project reinve
Poor audit trails	Spreadsheet tracking fails reg
High failure cost	Data errors cause operational

These challenges are particularly acute in Financial Services, where a single data migration error can trigger regulatory penalties, operational disruptions, and significant financial loss.

Our Solution

The FS Migration Validation Engine provides:

1. Automated Schema Discovery -- Connect to source and target databases, discover schemas, extract metadata

2. 10 Structured Validation Controls -- Comprehensive validation framework covering row counts, data types, nullability, keys, referential integrity, business rules, date boundaries, numeric precision, string patterns, and regulatory completeness

3. Objective Quality Scoring -- Automated pass/fail scoring with configurable thresholds

4. Release Gate Governance -- Block migrations that fail to meet minimum quality standards

5. Audit-Ready Exports -- CSV audit trails suitable for regulatory review

6. REST API -- Programmatic integration with CI/CD pipelines and deployment workflows

Why Azure

Microsoft Azure provides the enterprise-grade foundation required for regulated Financial Services deployments:

Requirement	Azure Service
------------	---------------
Scalable compute	Azure Container Apps -- serverl
Managed database	Azure SQL Database -- high avai
Secure storage	Azure Blob Storage -- encrypted
Secrets management	Azure Key Vault -- no credentia
Identity & access	Microsoft Entra ID -- enterpris
API management	Azure API Management -- gateway
Monitoring	Azure Monitor + Application In
Security	Microsoft Defender for Cloud --

Architectural alignment:

- ? Azure Well-Architected Framework (all 5 pillars)
- ? Cloud Adoption Framework
- ? Zero Trust security model
- ? Infrastructure as Code (Bicep / ARM)
- ? CI/CD with GitHub Actions / Azure DevOps

Platform Architecture

```
Users / CLI / API
           ?
???????????????
? Platform    ?
? Core        ?
???????????????
           ?
?????????????????????????????
?          ?          ?          ?          ?
Discovery Validation Scoring Governance
```

The Platform Core orchestrates execution across independent domain engines. Each engine communicates only with the Platform Core, ensuring loose coupling, independent scalability, and centralised governance.

Current implementation status:

Component	Status	Location in Codebase
-----------	--------	---------------------
CLI Engine	? Production-ready	`app/main.py`, `app/execution_
10 Validation Controls	? Implemented	`sql/controls/`
Scoring Engine	? Implemented	`app/scoring_engine.py`
Release Gates	? Implemented	`config.yaml` -- minimum score
Audit Export	? Implemented	`app/audit_export.py` -- CSV ge
REST API	? Built	`app/api/` -- FastAPI endpoints
Docker Deployment	? Ready	`Dockerfile`, `docker-compose.
Database Schema	? Production-ready	`sql/schema/`

Security Strategy

Implemented according to Zero Trust principles:

- No secrets in code -- All credentials via environment variables -> Azure Key Vault
- Encryption at rest and in transit -- TLS 1.2+, Azure SQL TDE, Blob encryption
- Audit logging -- Every validation execution logged with batch ID and timestamp
- Failure isolation -- Control-level isolation prevents cascade failures
- Release gates -- Automated quality gates block non-compliant migrations

Competitive Advantages

Area	Our Advantage
------	---------------
Purpose-built	Designed for regulated Financi
10 structured controls	Comprehensive, standardised va
Governance-first	Release gates, audit trails, c
Objective scoring	Automated, repeatable quality
Docker-native	Deploy anywhere -- cloud, on-pr
API-first	Integrate with existing CI/CD
Open architecture	Python-based, extensible, plug

Scale & Performance

The platform is designed for enterprise-scale migration validation:

- Database discovery -- Connect to multiple source and target databases
- Batch processing -- Validate migrations in structured batch workflows
- Horizontal scaling -- Stateless API design, containerised deployment
- Control isolation -- Independent execution per control with failure isolation

- Configurable timeout -- Per-control timeout (default: 300 seconds)

Roadmap

Phase	Capabilities	Timeline
-------	-------------	----------
Phase 1	CLI engine, 10 controls, scori	? Complete
Phase 2	REST API, dashboard, AI-assist	? In Development
Phase 3	Multi-tenant SaaS, web portal,	? Planned
Phase 4	Partner integrations, Microsof	? Future

Value to Microsoft

This platform demonstrates effective use of the Microsoft ecosystem by:

- Building on Azure-native managed services (Container Apps, SQL, Blob, Key Vault)
- Following the Azure Well-Architected Framework across all 5 pillars
- Leveraging Microsoft Entra ID for enterprise identity (future)
- Supporting GitHub Actions / Azure DevOps for CI/CD
- Providing a scalable SaaS foundation suitable for Azure Marketplace
- Creating Azure consumption growth as customer adoption scales

As customer adoption grows, usage expands across Azure compute, storage, identity, AI, monitoring, and integration services -- reinforcing long-term engagement with the Microsoft cloud platform.

Conclusion

The FS Migration Validation Engine represents a modern approach to data migration assurance for regulated Financial Services. Its automated validation framework, metadata-driven architecture, and Azure-native design provide financial institutions with the confidence, governance, and auditability required for mission-critical data migrations.

By aligning with Microsoft's architectural guidance and leveraging Azure's managed services, the platform is positioned to deliver measurable value to enterprise customers while demonstrating strong technical alignment with the goals of the Azure Founders Hub program.

Phase 2.1 - Executive Pitch Narrative

Phase 2.1 -- Executive Pitch Narrative

Version: 2.1

Product: FS Migration Validation Engine

Target: Microsoft Founders Hub, Investors, Enterprise Customers

Currency: GBP (£) -- UK-based company

Vision

Financial Services institutions migrating mission-critical data between systems face a fundamental challenge: how to ensure accuracy, completeness, and regulatory compliance when millions of records are in motion.

Our vision is to make data migration validation automated, objective, and audit-ready -- transforming it from a manual, high-risk exercise into a repeatable, governed, and verifiable process.

The Problem

Data migration in Financial Services is broken:

- Manual validation -- Teams spend weeks manually sampling and checking data
- Incomplete coverage -- Sampling misses edge cases and exceptions
- No standardisation -- Every migration project reinvents validation from scratch
- Poor auditability -- Spreadsheets and emails fail regulatory scrutiny
- High cost of failure -- A single data error can cause operational losses, regulatory fines, and reputational damage

Banks migrating core banking systems, payment platforms, or regulatory reporting systems cannot afford data quality failures. Yet most validation today relies on custom scripts, manual checks, and hope.

Our Solution

The FS Migration Validation Engine is an automated, metadata-driven validation and governance platform that:

1. Discovers legacy and target system schemas automatically
2. Validates data across 10 structured migration controls
3. Scores migration quality with objective, repeatable metrics
4. Governs the migration lifecycle with release gates and audit trails
5. Reports comprehensive audit evidence suitable for regulatory review

How It Works

```
Legacy System -> Discovery -> 10 Validation Controls -> Scoring -> Release Gate -> Audit Repo
```

The platform runs as a CLI engine, REST API, or Docker container -- integrating into existing CI/CD pipelines and deployment workflows.

Why Now?

Several trends make this the right time for an automated migration validation platform:

Trend	Impact
-------	--------
Regulatory pressure	FCA, PRA, Basel IV, SOX demand
**Cloud migration acceleration	Banks are moving core systems
AI expectations	Institutions expect intelligen
Legacy system retirement	COBOL, mainframe, and legacy p
Cost pressure	Financial institutions need to

Why Azure?

Microsoft Azure provides the enterprise-grade foundation required for regulated Financial Services:

- Azure Container Apps -- Scalable, serverless compute
- Azure SQL Database -- Managed, highly available database
- Azure Key Vault -- Secure secrets management
- Microsoft Entra ID -- Enterprise identity and access
- Azure Monitor -- Comprehensive observability
- Azure OpenAI -- Future AI-assisted mapping capabilities

Competitive Advantage

Area	Our Advantage
------	---------------
Purpose-built	Designed for regulated Financi
10 structured controls	Comprehensive, standardised va
Governance-first	Release gates, audit trails, c
Objective scoring	Automated, repeatable quality
Docker-native	Deploy anywhere -- cloud, on-pr
API-first	Integrate with existing CI/CD

Business Model

The platform will be offered as a SaaS subscription with tiered pricing:

Tier	Target	Price Range (GBP/month)

------	--------	-----------------------
Starter	Small migrations, pilot projec	£1,500-£3,500
Professional	Mid-size bank migrations	£3,500-£10,000
Enterprise	Large-scale, multi-system migr	£10,000-£35,000
Strategic	Enterprise-wide deployment, cu	Custom pricing

Typical annual contract value: £30,000-£80,000 per customer

Additional revenue streams:

- Professional services (implementation, custom controls at £5k-£20k each)
- AI-assisted mapping module (future)
- Enterprise support and SLAs

Go-to-Market

Primary channels:

- Microsoft Founders Hub and Azure Marketplace
- Direct enterprise sales to UK Financial Services CTOs and Heads of Data
- System integrator partnerships (Accenture, Deloitte UK, Infosys UK)
- Microsoft partner network and co-sell programme

Target customers:

- Tier 1-3 Banks (Barclays, Lloyds, NatWest, challenger banks)
- Insurance companies (Aviva, Legal & General, AXA UK)
- Asset management firms
- FinTech and payment processors
- Regulated enterprises undergoing data migration

Product Roadmap

Phase	Capabilities	Timeline
-------	-------------	----------
Phase 1	CLI engine, 10 controls, scori	? Complete
Phase 2	REST API, dashboard, AI-assist	? In Development
Phase 3	Multi-tenant SaaS, web portal,	? Planned
Phase 4	Partner integrations, Microsof	? Future

Investment Opportunity

Azure Founders Hub support will accelerate:

- Product development -- Complete Phase 2 (API, dashboard, AI mapping)
- Customer validation -- Secure 5+ pilot customers in UK Financial Services
- Security certifications -- ISO 27001, SOC 2 readiness
- Marketplace readiness -- Azure Marketplace listing

- Commercial growth -- Early customer acquisition targeting £450k ARR

Funding ask for Founders Hub: Azure credits (£120k max) + partner programme support

Seed round (if required): £500,000-£750,000

The platform is designed to grow Azure consumption as customers scale their migration initiatives, creating long-term value for both customers and the Microsoft ecosystem.

Closing Statement

Our mission is to transform data migration validation from a slow, manual, high-risk exercise into an automated, governed, and audit-ready process. By combining Azure-native architecture with intelligent automation, we aim to become the standard platform for regulated data migration assurance -- helping Financial Services institutions migrate with confidence, compliance, and speed.

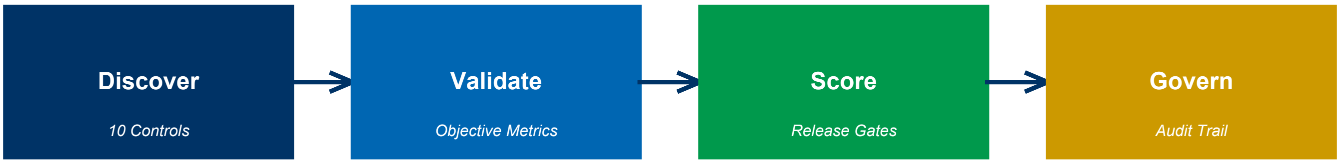

Figure 8: Business Flow - Discover, Validate, Score, Govern

Phase 2.2 - Investor Pitch Deck

Phase 2.2 -- Investor Pitch Deck

Version: 2.1

Product: FS Migration Validation Engine

Recommended Length: 14 Slides

Currency: GBP (£) -- UK-based company

Slide 1 -- Title

Data Migration Validation for Regulated Financial Services

FS Migration Validation Engine -- Automated, Governed, Audit-Ready

Built on Microsoft Azure

"Making data migration validation automated, objective, and compliant."

Slide 2 -- The Problem

Data Migration Validation Is Broken

Financial Services institutions rely on:

- ? Manual sampling (weeks of effort, misses edge cases)
- ? Spreadsheet tracking (fails regulatory scrutiny)
- ? Custom scripts (not repeatable, not standardised)
- ? Gut feel and hope (not objective, not auditable)

Consequences:

- Regulatory fines for data integrity failures
- Operational losses from undetected errors
- Migration delays and budget overruns
- Reputational damage

Slide 3 -- Market Opportunity

A Growing Regulatory Imperative

Target Market: Financial Services -- Banks, Insurers, Asset Managers, FinTech

Market Drivers:

- Regulatory pressure (FCA, PRA, Basel IV, SOX)

- Accelerated cloud migration in Financial Services
- Legacy system retirement (COBOL, mainframe)
- AI and automation expectations

Market Size: Global data migration market projected to grow significantly as Financial Services accelerates cloud adoption.

Slide 4 -- Our Solution

Automated Migration Validation Platform

The FS Migration Validation Engine provides:

? Automated schema discovery -- Connect, discover, and compare

? 10 structured validation controls -- Comprehensive, standardised

? Objective scoring -- Automated pass/fail with configurable thresholds

? Release gate governance -- Block bad migrations before they reach production

? Audit-ready exports -- CSV trails suitable for regulatory review

---

Slide 5 -- How It Works

Automated Validation Workflow

```
Legacy System
     ?
Schema Discovery (automated)
     ?
10 Validation Controls (C01-C010)
     ?
Objective Scoring (pass/fail per control)
     ?
Release Gate (min 80% score required)
     ?
Audit Report (CSV export)
```

The platform runs as CLI, REST API, or Docker -- integrating into any CI/CD pipeline.

Slide 6 -- Why We're Different

Feature	Manual / Custom	Our Platform
---------	---------------	--------------
Schema discovery	Manual analysis	? Automated
Validation coverage	Sampling (~20%)	? 100% of records
Scoring	Subjective	? Objective metrics
Governance	Spreadsheet tracking	? Release gates + audit
Repeatability	None (new scripts each time)	? Standardised framework

Regulatory audit	Manual evidence gathering	? Automated CSV export

Slide 7 -- Technology

Azure-Native Architecture

Built on Microsoft Azure:

- Azure Container Apps -- Serverless compute
- Azure SQL Database -- Managed, HA database
- Azure Key Vault -- Secrets management
- Microsoft Entra ID -- Enterprise IAM (future)
- Azure Monitor -- Observability
- Azure API Management -- API gateway

Architecture: Platform Core + microservices -- modular, extensible, cloud-native.

Slide 8 -- Business Model

SaaS Subscription Pricing

Tier	Target Customer	Price (GBP/month)
------	----------------	-------------------
Starter	Pilot projects, small migratio	£1,500-£3,500
Professional	Mid-size bank migrations	£3,500-£10,000
Enterprise	Large-scale migrations	£10,000-£35,000
Strategic	Enterprise-wide deployment	Custom

Additional: Professional services, AI mapping module (future), support/SLAs.

Slide 9 -- Go-to-Market

Channel Strategy

Primary:

- Microsoft Founders Hub -> Azure Marketplace
- Direct enterprise sales (CTOs, Heads of Data)
- System integrator partnerships (Accenture, Deloitte, Infosys)

Target Verticals:

- Tier 1-3 Banks
- Insurance companies
- Asset management
- FinTech / payments
- Regulated enterprises

Slide 10 -- Product Roadmap

Growth Plan

Phase	Capabilities	Status
-------	-------------	--------
Phase 1	CLI engine, 10 controls, scori	? Complete
Phase 2	REST API, dashboard, AI mappin	? In Development
Phase 3	Multi-tenant SaaS, web portal	? Planned
Phase 4	Marketplace, Fabric integratio	? Future

Slide 11 -- Why Microsoft

Built for Azure

Our platform increases Azure consumption through:

- Azure Container Apps
- Azure SQL Database
- Azure Blob Storage
- Azure Key Vault
- Azure OpenAI (future)
- Azure Monitor

Alignment:

- ? Azure Well-Architected Framework
- ? Cloud Adoption Framework
- ? Zero Trust security
- ? Marketplace readiness

Slide 12 -- Investment Opportunity

Use of Funds

Support from Founders Hub will accelerate:

Area	Allocation
------	-----------
Product development (Phase 2)	40%
AI mapping module (Azure OpenA	20%
Customer pilots & validation	20%
Security certifications	10%
Marketplace readiness	10%

Slide 13 -- Vision

The Future

Our goal is to become the standard platform for regulated data migration assurance.

Future capabilities:

- AI-assisted mapping recommendations
- Intelligent mapping engine with confidence scoring
- Interactive dashboards
- Regulatory template library (Basel, SOX, FCA, PRA)
- Multi-cloud support (AWS, GCP)

Slide 14 -- Closing

Migration Validation Starts with Certainty

"Don't guess your migration quality -- validate it."

FS Migration Validation Engine

Built on Microsoft Azure

Contact: [Your Contact Information]

Phase 2.3 - Market Analysis TAM/SAM/SOM

Phase 2.3 -- Market Analysis

TAM - SAM - SOM - Competitor Landscape

Version: 2.0

Product: FS Migration Validation Engine

Target: Financial Services Data Migration Validation Market

Executive Summary

The FS Migration Validation Engine addresses a growing and critical market: automated data migration validation for regulated Financial Services. As banks and financial institutions accelerate cloud migration, the need for automated, governed, and audit-ready validation becomes a regulatory and operational necessity.

The market is distinct from generic ETL, data quality, or testing tools -- it sits at the intersection of RegTech (Regulatory Technology) , data migration, and enterprise governance.

Market Drivers

Regulatory Pressure

Financial regulators globally are demanding demonstrable data integrity:

- FCA (UK) -- Operational resilience and data integrity requirements
- PRA (UK) -- Prudential regulation data standards
- Basel IV -- Enhanced data quality and reporting requirements
- SOX (US) -- Financial reporting data integrity
- GDPR -- Data protection during migration

Cloud Migration Acceleration

Financial Services cloud adoption is accelerating:

- 70%+ of banks have active cloud migration programmes
- Core banking system modernisation is a top 3 priority
- Regulatory reporting systems moving to cloud-native platforms

Legacy System Retirement

Critical legacy platforms are losing expert support:

- COBOL, mainframe, PowerBuilder, Oracle Forms
- Retiring workforce with platform-specific knowledge

- Urgent need for automated migration assurance

AI & Automation Expectations

Financial institutions expect intelligent automation:

- Manual validation is no longer acceptable
- AI-assisted mapping and validation is the expected standard
- Governance and audit must be automated, not manual

Target Market

Primary Customers

Segment	Examples	Pain Point
---------	----------	------------
Tier 1 Banks	HSBC, Barclays, Lloyds, NatWes	Multi-system migrations, regul
Tier 2-3 Banks	Metro Bank, Virgin Money, chal	Migration cost and risk reduct
Insurance	Aviva, Legal & General, AXA	Policy and claims data migrati
Asset Management	BlackRock, Schroders, Legal &	Fund data accuracy during migr
FinTech / Payments	Wise, Revolut, Checkout.com	Rapid migration cycles, compli
Regulated Enterprises	Healthcare, Government, Utilit	Data integrity obligations

Secondary Customers

Segment	Opportunity
---------	-------------
System Integrators	Accenture, Deloitte, Infosys --
Cloud Migration Practices	Microsoft consulting partners,
Audit & Compliance Firms	Big 4 audit firms -- use for mi

Total Addressable Market (TAM)

The TAM encompasses global spend on:

- Financial Services data migration projects
- RegTech solutions for data governance and compliance
- Enterprise data quality and validation tools
- Cloud migration assurance services

Estimated global TAM: $8-12 billion annually

Note: Independent analyst data (Gartner, IDC) should be sourced for final figures.

Serviceable Available Market (SAM)

The SAM focuses on organisations that:

- Operate within the Microsoft Azure ecosystem
- Are undertaking active data migration projects
- Have regulatory compliance requirements
- Prefer SaaS-based tooling over custom builds

Estimated SAM: $1.5-2.5 billion annually

Serviceable Obtainable Market (SOM)

The initial SOM targets:

- UK Financial Services institutions (50+ Tier 1-3 banks, insurers, asset managers)
- Microsoft Azure-committed organisations
- Early adopters of automated migration governance
- Phase 1: 10-20 enterprise customers

Estimated SOM (Year 1-2): $10-30 million annually

Competitive Landscape

Category 1: Manual / In-House Solutions

Description: Custom scripts, spreadsheets, manual sampling

Strengths: Familiar, no licensing cost

Weaknesses: Not scalable, no governance, error-prone, no audit trail

Our Advantage: Automation, repeatability, governance, audit readiness

Category 2: ETL Testing Tools

Examples: QuerySurge, iCEDQ, Datagaps

Strengths: Data comparison capabilities

Weaknesses: Generic, not purpose-built for migration, limited governance

Our Advantage: Migration-specific controls, release gates, financial services focus

How They Compare:

Capability	Our Platform	ETL Testing Tools	Manual
----------	------------	------------------	--------
Purpose-built for migration	? Yes	? Generic testing	?
10 structured controls	? Built-in	?? Partial	?
Release gate governance	? Built-in	?	?
Audit trail	? Automated	?? Limited	?
Financial Services focus	? Purpose-built	?	?
Docker deployment	? Ready	?	N/A
API-first	? Built	?? Partial	?

Category 3: Data Quality Platforms

Examples: Informatica, Talend, Ataccama

Strengths: Broad data quality capabilities

Weaknesses: Complex, expensive, not migration-specific, long implementation

Our Advantage: Migration-focused, lightweight deployment, faster time-to-value

Category 4: Big 4 & SI Migration Practices

Examples: Accenture Migration Factory, Deloitte Migration Services

Strengths: Domain expertise, end-to-end service

Weaknesses: High cost, manual-heavy, not repeatable across clients

Our Advantage: Automated platform, consistent methodology, lower cost

Category 5: Cloud-Native Migration Tools

Examples: Azure Migrate, AWS Migration Hub, Google Cloud Migration

Strengths: Cloud infrastructure integration

Weaknesses: Infrastructure-focused, limited data validation, no governance

Our Advantage: Data-level validation, governance, Financial Services compliance

---

Competitive Positioning

Our platform occupies a unique position at the intersection of:

- RegTech -- Regulatory compliance for data migration
- Data Validation -- Automated, comprehensive, repeatable
- Migration Governance -- Release gates, audit, compliance
- Azure-Native -- Purpose-built for Microsoft ecosystem

Key message: Not a generic testing tool -- a purpose-built Migration Assurance Platform for regulated Financial Services.

Market Entry Strategy

Phase	Focus	Activities
-------	-------	------------
Year 1	UK Financial Services	Microsoft Founders Hub, pilot
Year 2	UK + Europe	Azure Marketplace, SI partners
Year 3	North America	Microsoft co-sell, multi-regio
Year 4+	Global	Marketplace ecosystem, regulat

Growth Opportunities

Opportunity	Description
-------------	-------------
**Regulatory template library*	Pre-built controls for Basel,
AI-assisted mapping	Azure OpenAI-powered schema ma
Multi-cloud validation	Extend to AWS, GCP migrations
Dashboard & reporting	Interactive visualisation of m
Managed migration service	Platform + implementation cons
Partner ecosystem	SI embed, marketplace, co-sell

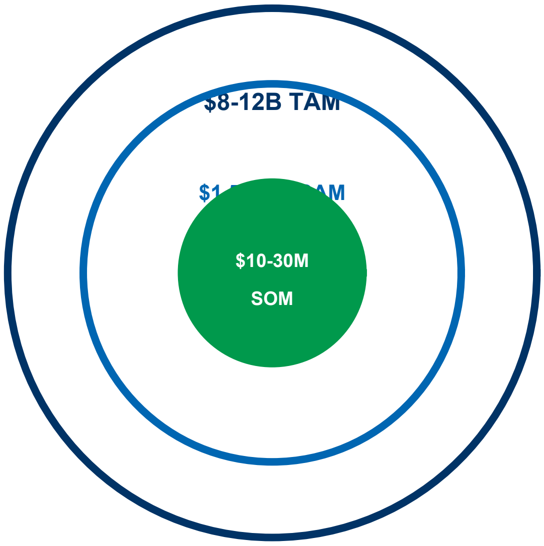

Figure 9: TAM / SAM / SOM Market Sizing

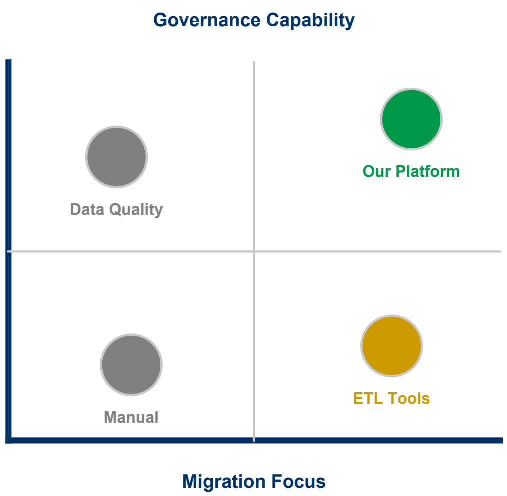

Figure 10: Competitive Positioning Quadrant

Phase 2.4 - Business Model and Pricing Strategy

Phase 2.4 -- Business Model & Pricing Strategy

Version: 2.1

Product: FS Migration Validation Engine

Currency: GBP (£) -- UK-based company

Executive Summary

The FS Migration Validation Engine follows a SaaS subscription model with tiered pricing based on migration project volume, number of systems, and support level. The model is designed for predictable recurring revenue (ARR) while allowing expansion through professional services and future AI modules.

Revenue is generated from four complementary streams:

1. Enterprise SaaS subscriptions -- Core recurring revenue
2. Professional services -- Implementation, custom controls, migration advisory
3. Enterprise support & SLAs -- Premium support tiers (future)
4. AI mapping module -- Azure OpenAI-powered recommendations (future)

Revenue Streams

1. SaaS Subscriptions (Primary)

Tier	Target	Price (GBP/month)	Includes
------	--------	-------------------	----------
Starter	Pilot projects, small migratio	£1,500-£3,500	1 migration project, 2 databas
Professional	Mid-size bank migrations	£3,500-£10,000	3 concurrent projects, 5 datab
Enterprise	Large-scale migrations	£10,000-£35,000	Unlimited projects, multi-data
Strategic	Enterprise-wide deployment	Custom	Custom controls, multi-region,

Typical annual contract value (ACV): £30,000-£80,000 per customer -- consistent with UK enterprise regtech SaaS benchmarks.

2. Professional Services

Service	Description	Price (GBP)
---------	-------------	-------------
Implementation	Platform setup, CI/CD integrat	£15,000-£40,000
Custom Controls	Development of customer-specif	£5,000-£20,000 per control
Migration Advisory	Migration validation planning	£1,500-£7,500/day

Professional services are typically 15-20% of total revenue in Years 1-2, reducing to 5-10% as the platform scales --

consistent with SaaS industry benchmarks.

3. Future Revenue Streams

Stream	Description	Expected Launch
--------	-------------	-----------------
AI Mapping Module	Azure OpenAI-powered schema ma	Phase 2 (Year 2)
Enterprise Support & SLA	24/7 support, guaranteed SLAs,	Phase 3 (Year 2-3)
Regulatory Templates	Pre-built controls for specifi	Phase 3 (Year 3)

Revenue Model Rationale

The pricing model is based on:

- Value-based pricing -- Priced on migration project value (typically £500k-£5M per programme), not per-record. A validation platf
- Tiered progression -- Clear upgrade path from pilot (£1.5k/mo) to enterprise-scale (£35k+/mo) as customers expand usage.
- Subscription model -- Predictable ARR with annual contracts (typical UK enterprise procurement).
- Low upfront, high lifetime -- Low barrier to entry (£1.5k/mo starter), high expansion revenue as customers validate more migrati

Unit Economics (Illustrative)

Metric	Year 1	Year 2	Year 3
--------	--------	--------	--------
Customers	10	40	120
Average ARR/Customer	£45,000	£60,000	£70,000
ARR	£450,000	£2,400,000	£8,400,000
Gross Margin	78%	82%	85%
CAC	£28,000	£22,000	£18,000
LTV (avg 4 years)	£180,000	£240,000	£280,000
LTV:CAC	6:1	11:1	16:1

Notes:

- *CAC includes founder-led sales time (Year 1) and dedicated sales resource (Years 2-3)*
- *LTV:CAC ratio of 6:1 in Year 1 is healthy for enterprise SaaS; improves to 16:1 at scale*
- *Churn expected at <10% in Year 1, declining to <5% by Year 3 -- consistent with enterprise regtech benchmarks*

Azure Infrastructure Cost Model

Service	Monthly (GBP) -- MVP	Monthly (GBP) -- Scale
---------	--------------------	---------------------
Azure Container Apps	£150-£400	£800-£2,500
Azure SQL Database	£120-£250	£400-£1,200
Azure Blob Storage	£40-£80	£150-£400
Azure API Management	£80-£150	£250-£500
Azure Key Vault	£8-£15	£40-£80

Azure Monitor	£40-£80	£150-£400
Total per customer	**£438-£975**	**£1,790-£5,080**

As the platform moves to multi-tenant SaaS, infrastructure costs per customer reduce significantly through shared tenancy.

Pricing Principles

Principle	Rationale
-----------	-----------
Value-based	Priced on migration project va
Tiered	Clear upgrade path as customer
Subscription	Predictable recurring revenue
**Enterprise annual contracts*	Standard UK enterprise procure
Low entry, high expansion	Starter tier for pilots, expan

UK Market Context

This pricing reflects UK Financial Services market conditions:

- UK banks typically spend £500k-£5M per migration programme
- UK enterprise regtech SaaS pricing ranges £30k-£150k ACV
- UK investors expect GBP-denominated projections with realistic UK salary costs
- UK corporate tax rate: 25% (2024-25)
- R&D tax credits available for software development (up to 27% of qualifying costs)
- Azure Founders Hub provides up to £120,000 in free Azure credits (GBP equivalent of $150k)

Azure Marketplace Strategy

The platform will be available through:

- Azure Marketplace -- Transactable SaaS offer (GBP pricing)
- Microsoft Co-Sell -- Eligible for Microsoft field seller incentives
- Private Offers -- Custom pricing for large enterprise deals

This aligns with Microsoft's partner ecosystem and provides UK Financial Services customers with Azure consumption commitment benefits.

Phase 2.5 - Go-to-Market Strategy and Financial Projections

Phase 2.5 -- Go-to-Market Strategy & 3-Year Financial Projectio

Version: 2.1

Product: FS Migration Validation Engine

Currency: GBP (£) -- UK-based company

Go-to-Market Strategy

Target Customer Segments

Primary (Year 1-2):

Segment	Target	Entry Strategy
---------	--------	----------------
Tier 2-3 Banks	UK challenger banks, building	Direct founder-led sales, pilo
Insurance	UK insurers with migration pro	Microsoft partner referrals
FinTech	Payment processors, digital ba	Azure Marketplace listing

Secondary (Year 2-3):

Segment	Target	Entry Strategy
---------	--------	----------------
Tier 1 Banks	Barclays, Lloyds, HSBC, NatWes	SI partnerships, longer sales
Asset Management	Fund managers, wealth platform	Industry events, case studies
Regulated Enterprises	Healthcare, Government	Compliance-led positioning

Go-to-Market Channels

Channel	Year 1 Contribution	Year 2 Contribution	Year 3 Contribution
---------	---------------------	---------------------	---------------------
Direct Sales (founder-led)	60%	40%	25%
Azure Marketplace	20%	25%	30%
System Integrators	10%	20%	25%
Microsoft Co-Sell	5%	10%	15%
Inbound / Content	5%	5%	5%

Sales Cycle

Segment	Cycle Length	Key Decision Makers
---------	-------------	-------------------
Small/Medium Banks	2-4 months	CTO, Head of Data, Programme D
Large Banks	6-12 months	CIO, CRO, Head of Migration
Insurance	3-6 months	CTO, Head of Transformation

FinTech	1-3 months	CTO, VP Engineering

Customer Acquisition Strategy

Year 1:

- Secure 3-5 pilot customers in UK Financial Services
- Build case studies and reference accounts
- Establish Azure Marketplace presence
- Leverage Azure Founders Hub credits (£120k) for infrastructure

Year 2:

- Scale to 20-30 customers across UK + Europe
- Establish SI partnerships (2-3 partners -- Accenture, Deloitte UK, Infosys UK)
- ISO 27001 certification

Year 3:

- 80-100 customers globally
- Microsoft co-sell programme participation
- North America market entry (with USD pricing option)

3-Year Financial Projections

Revenue Forecast

Metric	Year 1	Year 2	Year 3
--------	--------	--------	--------
Customers	10	40	120
Average ARR/Customer	£45,000	£60,000	£70,000
SaaS Revenue	£450,000	£2,400,000	£8,400,000
Professional Services	£100,000	£300,000	£500,000
Total Revenue	**£550,000**	**£2,700,000**	**£8,900,000**

Revenue assumptions:

- *Year 1: 10 customers at £3,750/mo average (£45k ARR) + implementation fees*
- *Year 2: 40 customers at £5,000/mo average (£60k ARR) -- expansion revenue from existing customers via up-sell*
- *Year 3: 120 customers at £5,833/mo average (£70k ARR) -- mix of new logos and expansion*
- *Professional services: 1 implementation per new customer (£15k-£30k each)*
- *Churn: 10% Year 1 -> 8% Year 2 -> 5% Year 3*

Cost of Sales

Category	Year 1	Year 2	Year 3
----------	--------	--------	--------
Azure Infrastructure	£60,000	£240,000	£720,000
Customer Support (1 FTE)	£40,000	£80,000	£200,000
Third-party Licences	£20,000	£50,000	£100,000
Total CoS	**£120,000**	**£370,000**	**£1,020,000**
Gross Margin	**78%**	**86%**	**89%**

Cost assumptions:

- *Azure costs reduce per-customer as shared tenancy scales*
- *Support: 0.5 FTE in Year 1 -> 2 FTE by Year 3*
- *Third-party: GitHub, monitoring, security scanning tools*

Operating Expenses

Category	Year 1	Year 2	Year 3
----------	--------	--------	--------
Salaries & Benefits			
Founder/CEO	£60,000	£70,000	£80,000
Senior Engineer (Python/Azure)	£90,000	£95,000	£100,000
Junior Engineer	£50,000	£55,000	£60,000
Sales/Business Development	--	£60,000	£80,000
Customer Success	--	--	£55,000
Total Salaries	*£200,000*	*£280,000*	*£375,000*
Employer NI & Pension (est. 20	£40,000	£56,000	£75,000
Total Staff Costs	**£240,000**	**£336,000**	**£450,000**
Other OpEx			
Office & Equipment	£30,000	£40,000	£60,000
Sales & Marketing	£40,000	£100,000	£200,000
Professional Advisors (legal,	£25,000	£35,000	£50,000
Insurance & Compliance	£15,000	£25,000	£40,000
Travel & Events	£10,000	£30,000	£50,000
R&D (non-staff)	£20,000	£30,000	£50,000
Total Other OpEx	**£140,000**	**£260,000**	**£450,000**
Total Operating Expenses	**£380,000**	**£596,000**	**£900,000**

Salary assumptions reflect UK market rates (outside London):

- *Senior Python/Azure engineer: £85k-£100k typical for UK startup/FInTech*
- *Junior engineer: £45k-£55k*
- *Sales: £55k-£70k base + commission*
- *Employer NI (13.8%) + pension (3-5%) estimated at 20% on total salary*

EBITDA

Year	Revenue	Total Costs (CoS + OpEx)	EBITDA
------	---------	-------------------------	--------
Year 1	£550,000	£500,000	**£50,000**
Year 2	£2,700,000	£966,000	**£1,734,000**
Year 3	£8,900,000	£1,920,000	**£6,980,000**

Year 1 reaches break-even with £50k EBITDA -- achievable due to founder-led sales, low overhead, and Azure Founders Hub credits covering infrastructure.

Key Metrics

KPI	Year 1	Year 2	Year 3	Benchmark
-----	--------	--------	--------	-----------
ARR	£450,000	£2,400,000	£8,400,000	UK regtech median: £500k->£3M->£
Gross Margin	78%	86%	89%	SaaS target: >70% Year 1, >80%
Net Revenue Retention	105%	115%	125%	Enterprise SaaS median: 110%+
Customer Churn	<10%	<8%	<5%	Enterprise target: <10% annual
CAC	£28,000	£22,000	£18,000	UK enterprise SaaS: £15k-£35k
LTV	£180,000	£240,000	£280,000	Based on 4-year avg customer l
LTV:CAC	6:1	11:1	16:1	Healthy SaaS: >3:1, Great: >10
EBITDA Margin	9%	64%	78%	UK SaaS Rule of 40 target
Headcount	3	5	8	Capital-efficient -- typical UK

Funding Requirement

Target: £500,000-£750,000 Seed Round (optional -- can bootstrap if Founders Hub credits sufficient)

Use of Funds:

Area	Allocation	Amount (GBP)
------	-----------	--------------
Product Development (Phase 2)	40%	£200,000-£300,000
AI Mapping Module (Azure OpenA	20%	£100,000-£150,000
Customer Pilots & Validation	20%	£100,000-£150,000
Security Certifications (ISO 2	10%	£50,000-£75,000
Marketplace & Sales Readiness	10%	£50,000-£75,000

Alternative Path: Azure Founders Hub provides up to £120,000 in Azure credits. Combined with founder funding and early customer revenue, the business can reach breakeven without external equity in Year 1.

UK-specific funding options:

- Azure Founders Hub -- £120k Azure credits + support
- Innovate UK Smart Grants -- Up to £500k for R&D projects
- UK Enterprise Investment Scheme (EIS) -- Tax relief for angel investors (30% income tax relief)
- British Business Bank Start Up Loans -- Up to £25k per founder

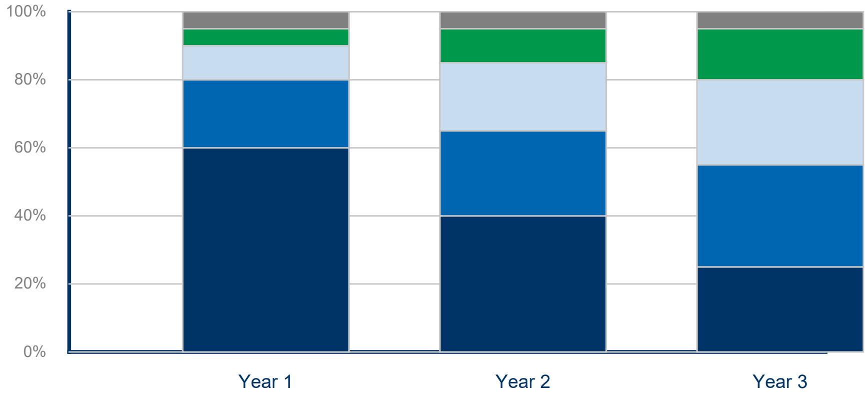

Figure 11: Channel Contribution by Year

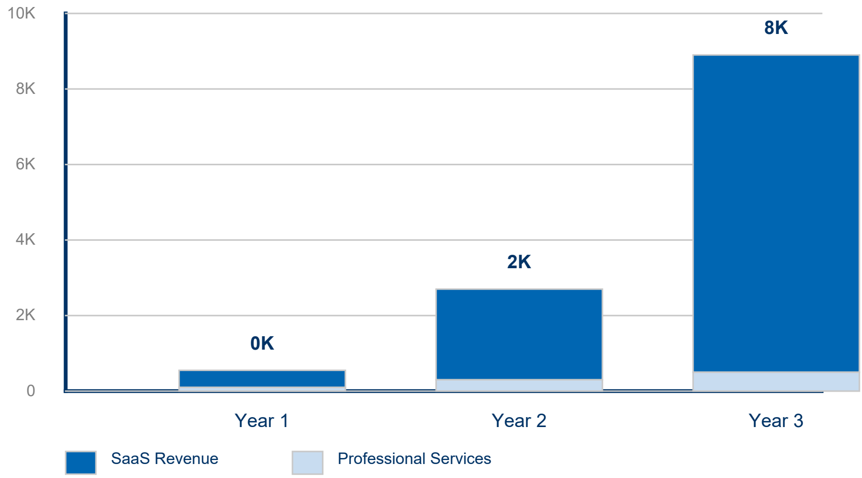

Figure 12: Revenue Growth - 3-Year Projection

Phase 2.6 - Financial Model and Funding Strategy

Phase 2.6 -- Financial Model & Funding Strategy

Version: 2.1

Product: FS Migration Validation Engine

Currency: GBP (£) -- UK-based company

Financial Model Summary

3-Year P&L

Item	Year 1	Year 2	Year 3
------	--------	--------	--------
Customers	10	40	120
SaaS Revenue	£450,000	£2,400,000	£8,400,000
Professional Services	£100,000	£300,000	£500,000
Total Revenue	**£550,000**	**£2,700,000**	**£8,900,000**
Cost of Sales	(£120,000)	(£370,000)	(£1,020,000)
Gross Profit	**£430,000**	**£2,330,000**	**£7,880,000**
OpEx (Staff + Other)	(£380,000)	(£596,000)	(£900,000)
EBITDA	**£50,000**	**£1,734,000**	**£6,980,000**
EBITDA Margin	9%	64%	78%

The business reaches EBITDA-positive in Year 1 -- achievable through low overhead, founder-led operations, and Azure Founders Hub credits (£120k) covering infrastructure costs.

Revenue Build-Up

Component	Year 1	Year 2	Year 3	Assumption
-----------	--------	--------	--------	------------
New customers	10	30	80	Growth driven by founder-led s
Retained customers	--	9	37	90% retention Year 1 -> 92% Yea
Total customers (EoY)	10	40	120	Net growth after churn
Average ARR/customer	£45,000	£60,000	£70,000	Expansion via up-sell + new lo
SaaS ARR	£450,000	£2,400,000	£8,400,000	Recurring subscription revenue
Professional services	£100,000	£300,000	£500,000	Implementation + custom contro

Azure Infrastructure Cost Model

Service	Monthly (MVP)	Monthly (Scale)
---------	--------------	-----------------
Azure Container Apps	£150-£400	£800-£2,500
Azure SQL Database	£120-£250	£400-£1,200

Azure Blob Storage	£40-£80	£150-£400
Azure API Management	£80-£150	£250-£500
Azure Key Vault	£8-£15	£40-£80
Azure Monitor	£40-£80	£150-£400
Monthly Total	**£438-£975**	**£1,790-£5,080**
Annual Total per customer	**£5,256-£11,700**	**£21,480-£60,960**

Year 1: Single-tenant deployment costs ~£5k/customer/year.

Year 3: Multi-tenant SaaS model reduces per-customer costs to £1k-£2k/customer/year.

Hiring Plan

Year	Role	Salary (GBP)	Notes
------	------	-------------	-------
Y1	Founder/CEO	£60,000	UK founder salary, below marke
Y1	Senior Engineer (Python/Azure)	£90,000	UK market rate for senior engi
Y1	Junior Engineer	£50,000	UK graduate + 1-2 years experi
Y2	Sales/Business Development	£60,000 + commission	Target for 30 new customers
Y3	Customer Success	£55,000	Support 120 customer accounts
Y3	Additional Engineer	£65,000	Scale development capacity

All salaries based on UK market rates (outside London). Employer NI (13.8%) + pension (3-5%) = ~20% on top.

Funding Strategy

Azure Founders Hub (Primary -- Year 1)

Benefit	Value (GBP)
---------	-------------
Free Azure credits	Up to £120,000
Microsoft partner enrolment	Free
Azure Marketplace listing	Included
Technical support	Included
Go-to-market support	Included

The Founders Hub credits alone cover Year 1 Azure infrastructure (£60k), with remaining credits available for development and testing environments.

Seed Round (If Required)

Item	Detail
------	--------
Target	£500,000-£750,000
Valuation basis	UK angel/Seed typical: £1.5M-£
Use	Product development, AI module
Timing	End of Year 1, after Founders
Investor target	UK angel networks, EIS-qualifi

Future Funding

Round	Target	Timing	Trigger
-------	--------	--------	---------
Seed	£500k-£750k	Year 1-2	Product validation, 10+ custom
Series A	£3M-£5M	Year 3	£2M+ ARR, proven GTM, SI partn

UK-Specific Funding Options

Source	Amount	Eligibility
--------	--------	-------------
Azure Founders Hub	Up to £120k credits	Startup stage
Innovate UK Smart Grant	Up to £500k	R&D projects, competitive appl
**EIS (Enterprise Investment S	Tax relief for investors	Must be EIS-qualified company
UK SEIS	Up to £250k investment	Early-stage, less than 2 years
British Business Bank	Various	Startup loans, growth guarante

Sensitivity Analysis

Scenario: Slower Customer Acquisition

Metric	Base Case	Downside (-30%)	Upside (+30%)
--------	-----------	-----------------	---------------
Year 1 customers	10	7	13
Year 1 revenue	£550,000	£385,000	£715,000
Year 1 EBITDA	£50,000	(£115,000)	£215,000
Year 3 ARR	£8.4M	£5.9M	£10.9M
Breakeven	Year 1	Year 2 (Q3)	Year 1

The business model is resilient -- even at 30% lower customer acquisition, breakeven is achieved by mid-Year 2 due to low fixed costs and variable infrastructure spend.

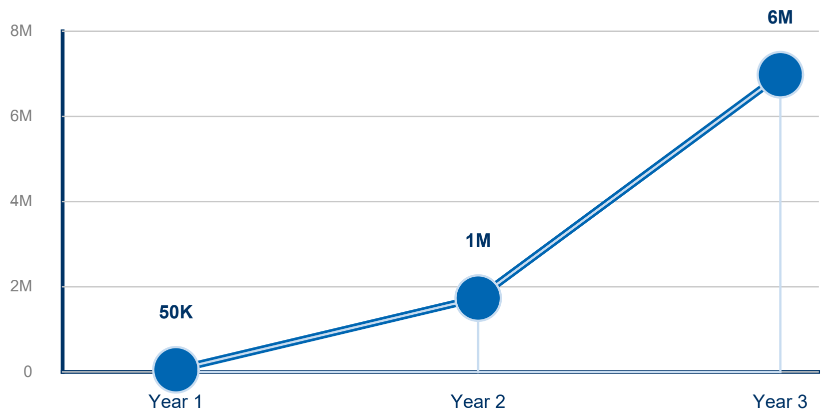

Figure 13: EBITDA Trajectory

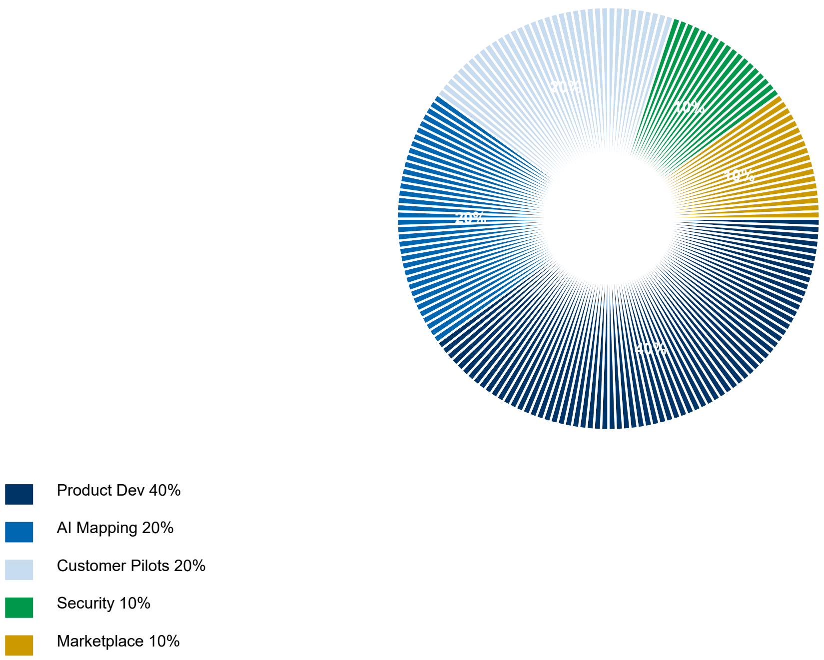

Figure 14: Use of Funds - Seed Round Allocation

Phase 2.7 - Competitive Differentiation

Phase 2.7 -- Competitive Differentiation

Version: 2.0

Product: FS Migration Validation Engine

Competitive Landscape

Category	Typical Vendors	Strengths	Weaknesses	Our Advantage
----------	----------------	-----------	------------	---------------
Manual / In-House	Custom scripts, spreadsheets	Familiar, no license cost	Not scalable, no governance	Automation, repeatability, aud
ETL Testing Tools	QuerySurge, iCEDQ, Datagaps	Data comparison	Not migration-focused, limited	Migration-specific controls, r
Data Quality Platforms	Informatica, Talend, Ataccama	Broad data quality	Complex, expensive, long imple	Lightweight, migration-focused
Big 4 / SI Services	Accenture, Deloitte Migration	Domain expertise	High cost, manual-heavy, not r	Automated platform, consistent
Cloud Migration Tools	Azure Migrate, AWS Migration H	Infrastructure integration	Infrastructure-focused, limite	Data-level validation, governa

Differentiation Matrix

Capability	Our Platform	ETL Testing	Data Quality	Manual
-----------	-------------	-------------	--------------	--------
Purpose-built for migration	?	?	?	?
10 structured controls	?	??	??	?
Release gate governance	?	?	?	?
Audit trail (automated)	?	??	??	?
Financial Services focus	?	?	?	?
Docker deployment	?	?	??	N/A
API-first	?	??	??	?
Objective scoring	?	??	??	?

Core Differentiators

1. Purpose-Built for Migration

Not a general-purpose testing or data quality tool -- designed specifically for data migration validation with 10 structured controls that cover the complete migration lifecycle.

2. Governance-First Design

Release gates, audit trails, exception registers, and compliance reporting are built into every execution -- not added as afterthoughts.

3. Financial Services Focus

Purpose-built for regulated environments. Controls align with regulatory requirements (FCA, PRA, Basel, SOX).

4. Lightweight Deployment

Docker-native, deploy in minutes. No complex installation, no heavy infrastructure requirements.

5. API-First Architecture

Integrate with existing CI/CD pipelines, deployment workflows, and enterprise toolchains.

6. Objective, Repeatable Scoring

Automated scoring removes subjectivity. Same inputs always produce same outputs -- essential for regulated environments.

Why Customers Choose Our Platform

Customer Need	Manual Approach	Our Platform
---------------	----------------	--------------
Complete validation coverage	Sampling only (~20%)	100% of records
Regulatory audit evidence	Manual evidence gathering	Automated CSV export
Repeatable process	New scripts each migration	Standardised framework
Migration governance	Spreadsheet tracking	Release gates + audit
Objective quality metrics	Subjective assessment	Automated scoring
Fast deployment	Weeks of setup	Docker, minutes to deploy

Phase 2.8 - Product Roadmap

Phase 2.8 -- Product Roadmap

Version: 2.0

Product: FS Migration Validation Engine

Vision

To become the standard platform for regulated data migration assurance -- helping Financial Services institutions validate every data migration with automated controls, objective scoring, and auditable governance.

Product Evolution

```
MVP (CLI Engine)
    ?
    ?
Enterprise API Platform (REST API + Dashboard)
    ?
    ?
AI-Assisted Migration (Azure OpenAI Mapping)
    ?
    ?
Multi-Tenant SaaS (Web Portal + Marketplace)
    ?
    ?
Intelligent Migration Platform (AI Agents + Fabric)
```

Year 1 -- Foundation & Validation (Current)

Objective

Establish the platform with Financial Services customers and validate product-market fit.

Completed

Capability	Status
-----------	--------
CLI Engine (10 controls)	? Complete
PostgreSQL schema	? Complete
Scoring engine	? Complete
Release gates	? Complete
Audit export (CSV)	? Complete
Docker deployment	? Complete
Control dependency DAG	? Complete
Security scanning	? Complete

REST API (FastAPI)	? Built
Schema discovery	? Built
Rule executor	? Built

In Development

Capability	Timeline
-----------	----------
Web dashboard (interactive)	Q3
AI-assisted mapping (Azure Ope	Q4
Enterprise SSO (Entra ID)	Q4

Milestones

- ? 5 pilot customers in UK Financial Services
- ? Azure Marketplace listing
- ? ISO 27001 certification (Q4)
- ? First paid enterprise customer (Q3)

Year 2 -- Enterprise Platform

Objective

Expand to multi-tenant SaaS with enterprise-grade capabilities.

Capability	Description	Priority
-----------	-------------	----------
Multi-tenant architecture	Isolated customer workspaces	High
Interactive dashboard	Real-time migration quality vi	High
AI mapping recommendations	Azure OpenAI-powered schema ma	High
User management	RBAC, team collaboration	Medium
Advanced reporting	Custom report builder	Medium
API versioning	v2 API with enhanced capabilit	Medium
Power BI integration	Embedded analytics	Low

Milestones

- 30+ enterprise customers
- European market expansion
- 2+ system integrator partnerships
- Microsoft co-sell programme enrolment

Year 3 -- AI-Powered Migration Intelligence

Objective

Transition from automated validation to intelligent migration decision support.

Capability	Description

-----------	-------------
Intelligent mapping engine	ML-powered schema matching wit
Predictive risk scoring	AI models predict migration fa
Regulatory template library	Pre-built controls for Basel,
Anomaly detection	AI identifies unusual data pat
Natural language queries	Ask questions about migration
Automated remediation suggesti	AI recommends fixes for failed

Milestones

- 100+ customers
- $10M+ ARR
- North America market entry
- Series A fundraising

Year 4+ -- Intelligent Migration Platform

Objective

Become the comprehensive platform for migration governance and intelligence.

Capability	Description
-----------	-------------
AI agents	Autonomous migration monitorin
Microsoft Fabric integration	Enterprise data platform conne
Marketplace	Third-party control and integr
Multi-cloud	AWS, GCP migration validation
Compliance automation	Automated regulatory reporting
Migration knowledge graph	Cross-project migration intell

Technology Roadmap

Capability	Y1	Y2	Y3	Y4+
-----------	:--:	:--:	:--:	:--:
CLI Engine	?			
REST API	?			
PostgreSQL	?			
Docker	?			
Web Dashboard	?	?		
Azure OpenAI	?	?		
Multi-Tenant SaaS		?		
RBAC / SSO		?		
Power BI		?		
AI Mapping Engine		?	?	
Regulatory Templates			?	
Predictive Risk			?	
Microsoft Fabric				?

Multi-Cloud				?
AI Agents				?
Marketplace				?

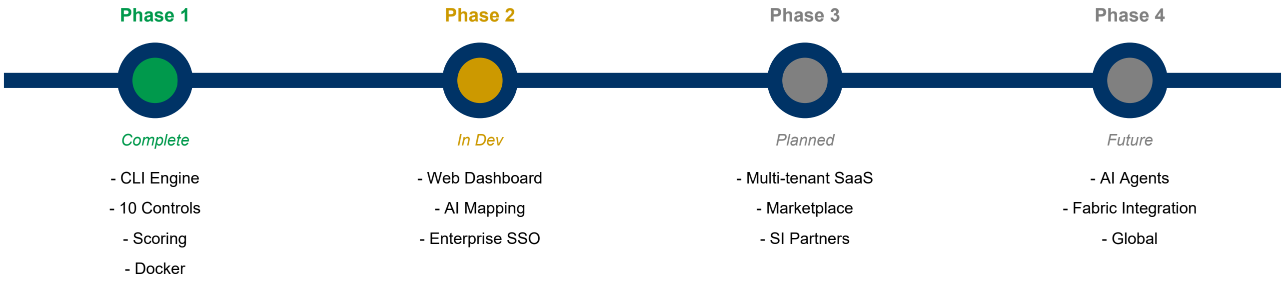

Figure 15: Product Roadmap Timeline

Phase 2.9 - Investor FAQ

Phase 2.9 -- Investor FAQ

Version: 2.1

Product: FS Migration Validation Engine

Currency: GBP (£) -- UK-based company

Company & Vision

1. What problem are you solving?

Financial Services institutions migrating data between systems lack automated validation tools. They rely on manual sampling, spreadsheets, and custom scripts -- resulting in incomplete coverage, no audit trail, and high risk of undetected data errors. In regulated environments, this is unacceptable.

2. Why is this problem important now?

Three trends converge: (1) accelerating cloud migration in Financial Services, (2) increasing regulatory pressure for data integrity (FCA, PRA, Basel IV, SOX), and (3) retirement of legacy platform experts. Institutions need automated, governed, and audit-ready validation -- not manual processes from a bygone era.

3. Who are your customers?

Tier 1-3 banks, insurance companies, asset managers, FinTechs, and regulated enterprises undergoing data migration. Initial focus on UK Financial Services, expanding to Europe and North America.

4. How large is the opportunity?

Global TAM estimated at $8-12B annually across Financial Services data migration, RegTech, and data validation. SAM of $1.5-2.5B for Azure-centric FSI organisations. SOM of $10-30M in Year 1-2.

Product

5. What does the platform do?

Automated data migration validation. It discovers source/target schemas, runs 10 structured validation controls (row counts, data types, keys, business rules, etc.), scores migration quality, enforces release gates, and produces audit-ready CSV exports.

6. What makes you different from ETL testing tools?

We're purpose-built for data migration, not generic ETL testing. Our 10 controls are standardised for migration validation. We include governance features (release gates, audit trails, dependency management) that ETL tools don't have. And we're focused specifically on regulated Financial Services.

7. Is your technology defensible?

Yes -- our defensibility comes from: (1) domain expertise in Financial Services migration, (2) proprietary control framework (C01-C010), (3) modular Platform Core architecture, (4) customer implementation knowledge, and (5) regulatory alignment that generic tools can't easily replicate.

8. Do you use AI?

Our current platform is rule-based and metadata-driven -- delivering reliable, deterministic validation. We plan to integrate Azure OpenAI for AI-assisted mapping recommendations in Phase 2. We prioritise accuracy and auditability over black-box AI.

Market

9. Who are your competitors?

Manual in-house solutions, ETL testing tools (QuerySurge, iCEDQ), data quality platforms (Informatica, Talend), Big 4 migration services (Accenture, Deloitte), and cloud migration tools (Azure Migrate). None offer our combination of migration-focused controls, governance, audit, and lightweight deployment.

10. Why will customers buy from you instead of building internally?

Building an equivalent platform requires expertise in migration validation, database internals, governance frameworks, and regulatory compliance -- plus months of development. Our platform is ready to deploy today, standardised, and continuously improved.

11. What is your pricing model?

SaaS subscription -- £1,500-£3,500/month (Starter), £3,500-£10,000/month (Professional), £10,000-£35,000/month (Enterprise), custom for Strategic deployments. Plus professional services for implementation (£15k-£40k) and custom controls (£5k-£20k each).

Business

12. What is your business model?

Recurring SaaS subscriptions with tiered pricing based on migration project volume and support level. Additional revenue from professional services and future AI modules.

13. What are your expected margins?

Gross margins of 80%+ as we scale, typical for SaaS platforms. Operating margins improve from Year 2 as development costs stabilise and revenue scales.

14. How will you acquire customers?

Founder-led direct sales to Financial Services CTOs and Heads of Data, Azure Marketplace listings, system integrator partnerships (Accenture, Deloitte, Infosys), and Microsoft co-sell programme.

15. What is your sales cycle?

2-4 months for mid-size banks and FinTech, 6-12 months for large Tier 1 banks. Pilot projects provide a faster entry path.

Funding

16. How much funding are you seeking?

£500,000-£750,000 seed round, or utilisation of Azure Founders Hub credits (up to £120,000) for initial infrastructure and development. As a UK-based company, we are also exploring Innovate UK Smart Grants and EIS-qualified angel investment.

17. How will the funds be used?

40% product development (Phase 2), 20% AI mapping module (Azure OpenAI), 20% customer pilots, 10% security certifications, 10% marketplace readiness.

18. What milestones will this funding achieve?

Product validation with 5+ Financial Services customers, Azure Marketplace listing, ISO 27001 certification, repeatable sales process, and £450k+ ARR.

Risks

19. What are the biggest risks?

Enterprise sales cycles are longer than consumer SaaS. Competition from established data quality vendors. Customer adoption of new validation methodologies. These are mitigated through pilot programmes, SI partnerships, and Financial Services domain focus.

20. What prevents Microsoft or a large vendor from copying you?

Large vendors move slowly in specialised niches. Our Financial Services domain expertise, migration-specific control framework, and customer implementation knowledge create a moat. We're not competing with Azure Migrate -- we complement it.

Phase 2.10 - Demo Script

Phase 2.10 -- Demo Script

Version: 2.0

Product: FS Migration Validation Engine

Duration: 10-15 minutes

Objective

Demonstrate how the FS Migration Validation Engine automates data migration validation for Financial Services -- from schema discovery through 10 validation controls, objective scoring, release gate enforcement, and audit-ready exports.

Demo Flow

Section	Duration
---------	----------
Introduction & Problem	1 min
Platform Overview	2 mins
Schema Discovery	2 mins
Validation Execution (C01-C010	3 mins
Scoring & Release Gates	2 mins
Audit Export	2 mins
API Integration	1 min
Closing & Business Value	2 mins
Total	**~15 minutes**

---

1. Opening (1 Minute)

> *"Financial Services institutions migrating data between systems face a critical challenge: how to ensure accuracy, completeness, a*

>

> *Traditional approaches rely on manual sampling, spreadsheets, and custom scripts -- which are slow, error-prone, and fail regulato*

>

> *The FS Migration Validation Engine automates this process -- providing complete validation coverage, objective scoring, governan*

---

2. Platform Overview (2 Minutes)

Show the CLI interface and explain the architecture:

```
FS Migration Validation Engine v1.4
??? 10 Migration Controls (C01-C010)
??? Automated Scoring Engine
??? Release Gate Enforcement
??? Audit Export (CSV)
??? REST API (FastAPI)
??? Docker Deployment
```

> *"The engine runs as a CLI tool, REST API, or Docker container. It connects to source and target databases, runs 10 standardised v*

---

3. Schema Discovery (2 Minutes)

Show the discovery command:

```
python app/main.py discover --config config.yaml
```

Highlight:

- Automated connection to source and target databases
- Schema metadata extraction
- Table, column, and relationship discovery

> *"The discovery engine automatically connects to source and target databases, extracts schema information, and builds a metadata*

---

4. Validation Execution (3 Minutes)

Show the run command:

```
python app/main.py run --config config.yaml
```

Walk through controls being executed:

Control	Description	Example Output
---------	-------------	----------------
C01	Row count reconciliation	"PASS -- Source: 1,234,567 vs T
C02	Data type validation	"PASS -- All column types match
C03	Nullability check	"PASS -- All constraints satisf
C04	Primary key integrity	"PASS -- No duplicates"
C05	Foreign key integrity	"PASS -- All references valid"
C06	Business rule validation	"3 WARNINGS -- See exception re
C07-C010	Additional controls	"PASS / WARN / FAIL per contro

> *"Each control runs independently with failure isolation. Results are captured with unique batch IDs for full traceability. The control o*

---

5. Scoring & Release Gates (2 Minutes)

Show the scoring output:

```
Batch: ae40b96c-20da-4972-bb29-bff3c2451ae0
Overall Score: 87/100
Controls Passed: 8/10
Controls Warning: 2/10
Controls Failed: 0/10

RELEASE GATE: PASSED (minimum 80% threshold met)
Migration: APPROVED
```

> *"The scoring engine produces objective, repeatable quality metrics. The release gate enforces a minimum quality threshold -- if the*

---

6. Audit Export (2 Minutes)

Show the export command:

```
python app/main.py export --config config.yaml --batch-id <id>
```

Show sample CSV output:

```
batch_id,control_id,status,score,exceptions,timestamp
ae40...,C01,PASS,100,0,2026-06-23 10:00:00
ae40...,C02,PASS,100,0,2026-06-23 10:00:05
ae40...,C06,WARN,70,3,2026-06-23 10:00:30
```

> *"Every validation execution produces a complete audit trail -- suitable for regulatory review. No manual evidence gathering required*

---

7. API Integration (1 Minute)

Show a sample API call:

```
curl -X POST https://api.example.com/api/v1/validate \
  -H "Content-Type: application/json" \
  -d '{"project_id": "ae40...", "controls": ["C01","C02"]}'
```

> *"The FastAPI REST API enables integration with existing CI/CD pipelines, deployment workflows, and enterprise toolchains."*

---

8. Closing Summary (2 Minutes)

> *"In just a few minutes, we've demonstrated:*

>

> *? **Automated schema discovery** -- No manual analysis required*
> *? **10 structured validation controls** -- Complete, standardised coverage*
> *? **Objective scoring** -- Repeatable quality metrics*
> *? **Release gate governance** -- Automated migration approval/blocking*
> *? **Audit-ready exports** -- CSV evidence for regulatory review*

>

> *The platform is Azure-native, Docker-deployable, and purpose-built for regulated Financial Services data migration.*

>

> *Our mission is to transform data migration validation from a manual, high-risk exercise into an automated, governed, and audit-rea*

Demo Environment Requirements

Item	Details
------	---------
Engine	Running instance of FS Migrati
Databases	Source and target PostgreSQL d
Configuration	`config.yaml` with DemoBank se
CLI	Terminal with Python environme
Sample Data	`sql/demo/` -- demo source and
Backup	`engine_backup.sql`, `source_b

Key Messages to Reinforce

1. Purpose-built for Financial Services -- Not a generic ETL tool
2. Automated and repeatable -- Same inputs, same outputs, every time
3. Governance-first -- Release gates, audit trails, compliance built-in
4. Lightweight deployment -- Docker-native, deploy in minutes
5. Azure-native -- Built for the Microsoft ecosystem
6. Enterprise-ready -- Secure, scalable, auditable